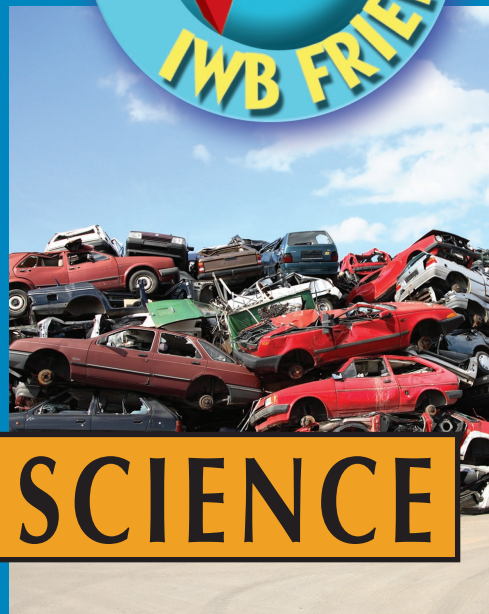


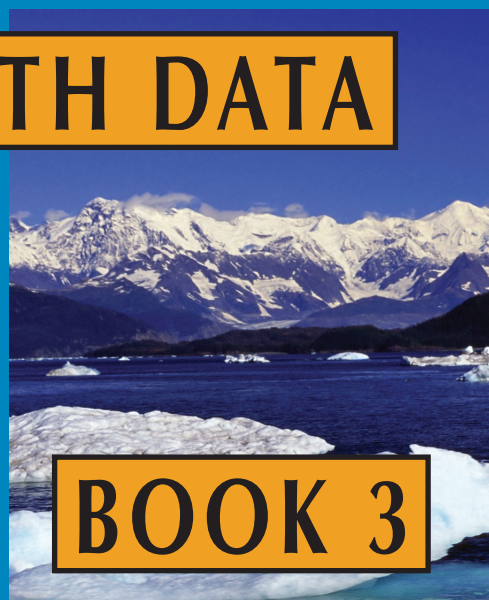
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THE SKILLS OF SCIENCE

WORKING WITH DATA



BOOK 3



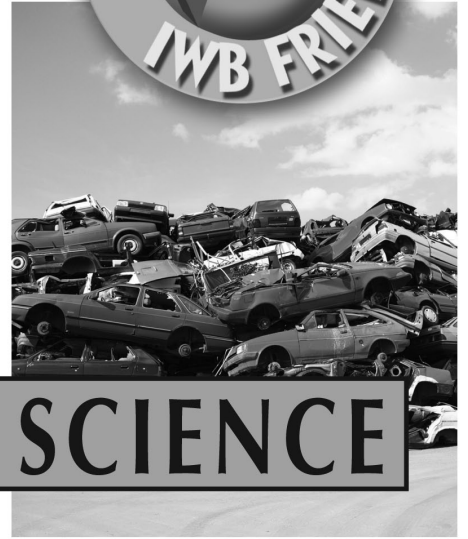
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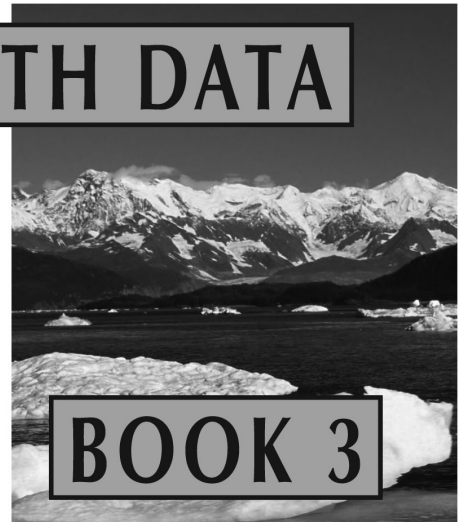
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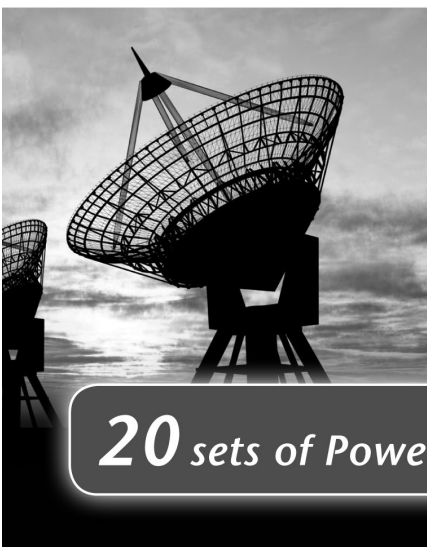


THE SKILLS OF SCIENCE



WORKING WITH DATA

BOOK 3



20 sets of PowerPoint slides

Sarah Sharp

Sarah Sharp has over 10 years' experience teaching Science and Computing Science in comprehensive and selective high schools. She has produced resources for the New South Wales Department of Education and Training on numeracy strategies and the use of dataloggers and their effective integration into teaching and learning programs. Sarah has led teacher workshops in using a variety of computer programs including dataloggers, constructed databases used in recording and analysing student performance and worked with industry providers to critically analyse the design and functionality of software for use in schools. Sarah has been a marker of the Essential Secondary Science Assessment (ESSA) and the School Certificate Science examination papers for several years and is passionate about educating people in science and implementing technology to enrich learning opportunities.

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The *Skills of Science* series is about the nature of science and testing ideas. It encourages an everyday perception of science, conceptual understanding and thinking skills, focusing on how students can learn to think and explore scientifically.

For each title in the series, there is an accompanying CD-ROM containing 20 PowerPoint presentations to summarise and extend the content explored in the topics in the book. These presentations are further complemented by a follow-up activity sheet based on each one, also on the CD-ROM.

The first two books of this series focus on the Scientific Method, and how and why it is used.

This third book in the series, *Working with data*, focuses on the collection, description and organisation of data obtained from experiments. It is divided into three content sections:

- **Collecting data** develops students' skills in designing experiments. Through these activities, students will explore the different types of laboratory equipment and the properties that make them appropriate for different tasks, the importance and the process of assessing the risk of any experiment, the accuracy and reliability of results, and the factors that contribute to the validity of data.
- **Describing data** explores appropriate ways of presenting measurements. In these activities there is a strong focus on fundamental mathematical skills and how they relate to scientific data. Students will investigate units of measurement, the use of scientific notation and the importance and application of significant figures. Three card games are provided for students to consolidate their understanding of orders of magnitude.
- **Organising data** investigates methods of collating data into logical formats. Through a series of practical exercises, students will present data in formats such as tables, timelines, models and databases. In this section, students will need access to a computer with PowerPoint (for topic 22) and a database program such as Access (for topics 24–26).

Book four of this series, *Dealing with graphs*, develops students' skills in graphing organised data to obtain meaning from their results. Book five, *Science writing made easy*, explores the effective synthesis and communication of the knowledge and understanding gained from experiments.

Topic	Title	Description	Page
Collecting data			
1	Drawing scientific diagrams	Rules for drawing scientific diagrams, identifying equipment and drawing apparatus	6
2	Choosing lab equipment A	Understanding how the properties of a material determine its use	8
3	Choosing lab equipment B	Understanding how the properties and design of lab equipment relate to function	9
4	Assessing risk	RIP scaffold for assessing risk in an experiment	11
5	Destructive and non-destructive testing	Distinguishing between destructive and non-destructive forms of testing	12
6	Reading scales	Stepped-out method for reading a scale	13
7	Quick quiz	Questions on topics 1 to 6	14
8	Experimental variables	Understanding independent, dependent and controlled variables	15
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11	Validity	Criteria for valid experiments	20
12	Quick quiz A	Questions on topics 8 to 11	22
Describing data			
13	Describing measurements A	Fundamental units of measurement, history of systems of measurement	23
14	Describing measurements B	Deriving units of measurement	24
15	Scientific notation	Writing numbers using scientific notation	25
16	Significant figures	Identifying the number of significant figures and rounding off to the correct number of significant figures after calculations	26
17	Orders of magnitude	Converting between scientific notation and large and small numbers; using prefixes for units	28
18	Orders of magnitude games	Three card games for consolidating understanding of orders of magnitude	30
19	Quick quiz B	Questions on topics 15 to 18	32
Organising data			
20	Drawing tables	Drawing well-organised tables for recording data	33
21	Drawing timelines	Rules for drawing timelines; history of the atomic model	35
22	Modelling using PowerPoint	Modelling a precipitation reaction using animation	37
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24	Using a database – designing	Creating a table and setting properties for fields	41
25	Using a database – entering data	Creating and using a form	43
26	Using a database – retrieving data	Creating and reporting on a query	44
27	Quick quiz	Questions about databases	45
Selected answers			46

What does it include?

The topics in this title are complemented by the information and activities contained on the CD-ROM at the back of this book.

The CD-ROM includes:

- 20 PowerPoint presentations to accompany the topics in this book, which summarise and extend the content.
- 20 follow-up activity sheets, each based on the matching PowerPoint presentation. These are supplied as PDF documents.

Note to the teacher

To get the most out of these teaching resources, teachers are encouraged to preview the PowerPoint presentations and accompanying activity sheets before using them in class.

Using the CD-ROM

In the Windows environment, the CD-ROM will autorun. From the Main menu, simply click the PowerPoint presentation you wish to view.

In the Macintosh environment, double-click the 'Science – Working with Data' icon. From the Main menu, click the PowerPoint presentation you wish to view.

You can also open each presentation from within the PowerPoint application.

The Slide Show view is used to display each PowerPoint presentation on the screen.

It is recommended that you use an external projector for whole-class use.

To access the activity sheets, simply click the pencil icon on the appropriate topic on the main menu.

Technical specifications

Windows

- The CD-ROM will autorun in the Windows environment, with an operating system of Windows XP or later.
- To edit a presentation, open it from within the PowerPoint application. Note that if editing these PowerPoint presentations in a version prior to Office 2000, some cosmetic losses may occur. These losses are cosmetic only, and will not affect the clarity and content of the slide show.

Macintosh

- Double-click the 'Science – Working with Data' icon on the CD-ROM. From the Main menu, click on the presentation you wish to view.
- To view a fully featured presentation, and edit it, you must have the PowerPoint 2004 application installed (i.e. Office 2004 for Macintosh or later).
- To edit a presentation, open it from within the PowerPoint application. Note that if editing these PowerPoint presentations in a Macintosh environment, some cosmetic losses may occur. These losses are cosmetic only, and will not affect the clarity and content of the slide show.

Scientists frequently use drawings or diagrams to communicate information. Since most scientists are not artists, scientific drawings of equipment are simple line diagrams. Scientific diagrams show how the equipment used to carry out experiments is set up, so that other scientists can repeat experiments in their own labs.

Several pieces of equipment set up to perform an experiment are called an *apparatus*.

- 1
- Unjumble the letters in brackets to complete the rules for drawing scientific diagrams.
- Rule 1: Always use a (ncliep) _____.

Rule 2: Draw all straight lines using a (lrrue) _____.

Rule 3: Make the drawing about 10 cm (gihh) _____.

Rule 4: The diagram should be a (wto) _____-dimensional cross section with no shading or colour.

Rule 5: All equipment should be (delbllae) _____ using straight, ruled lines with no arrowheads.

Rule 6: Label (iesnl) _____ should touch the equipment being named.

Rule 7: Equipment that is touching in real life should be shown to be (otuhgnc) _____ in the diagram; equipment does not float.

Rule 8: Make sure openings are drawn as open and are not shown with a (ild) _____.

Rule 9: Equipment should be drawn in proportion to other equipment. For example, a tiny (neBusn) _____ burner should not be shown heating a giant (erkabe) _____.

2

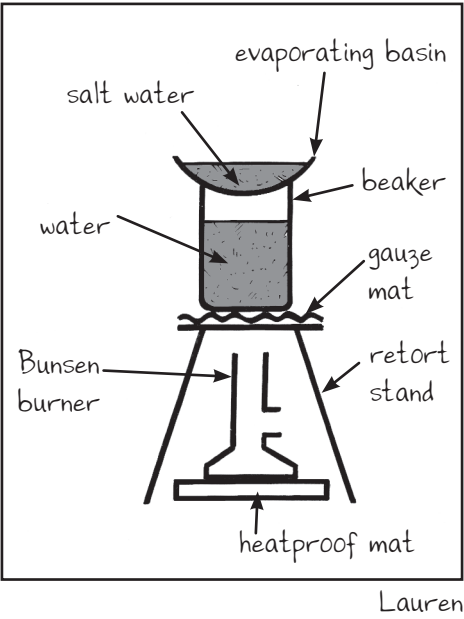
Use the word bank to write the correct name under each piece of scientifically drawn equipment.

WORD BANK			
• Bunsen burner	• heatproof mat	• thermometer	• filter funnel with filter paper
• retort stand with boss head and clamp	• gauze mat	• test tube in a test tube rack	• measuring cylinder
	• tripod	• evaporating basin	• conical flask
	• beaker		

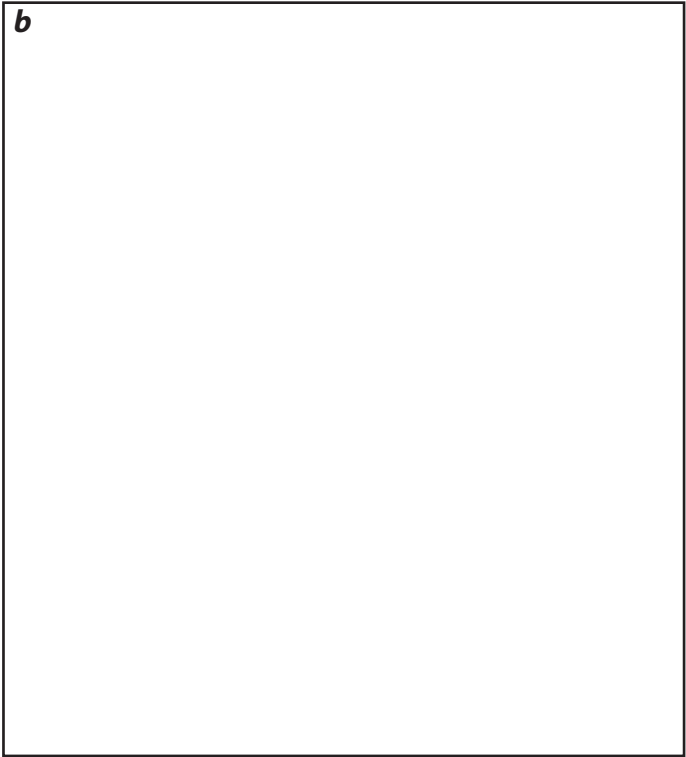
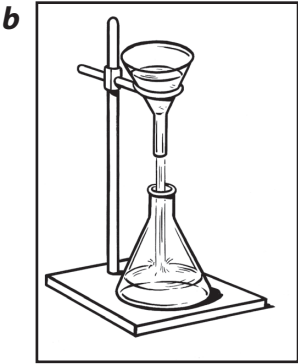
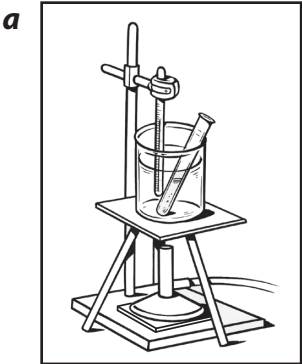
a	b	c	d
e	f	g	h
i	j	k	l

3 Lauren drew a scientific diagram of the apparatus she used to separate salt from water by evaporation. Use the marking criteria in the table below to give Lauren a mark for her diagram.

CRITERIA	CORRECT?
Pencil used	✓
Straight lines drawn using a ruler	
All equipment correctly labelled	
No floating equipment	
About 10 cm high	
Equipment is in correct proportion to each other	
Diagram is 2D, with no shading	
No arrowheads on label lines	
Label lines touch equipment	
Openings are not closed off	
Total	/ 10



4 Draw a scientific diagram of each apparatus shown below. Use the diagrams of the equipment in question 2 to help you.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Using incorrect equipment in a laboratory can be dangerous. The properties of the material from which the equipment is made is one criterion scientists use to choose an item of equipment.

A material's properties include such things as its strength, colour, melting point and density. Different properties are suitable for different functions. The function of a piece of equipment is what it is used for.

Brittle means it shatters if you drop it.

- 1 Identify two other properties of materials. _____
- 2 Listed below are some materials commonly used to make laboratory equipment. Write complete sentences below which match each material to the correct properties, function and example of use.

Materials

- Metals
- Porcelain, a type of ceramic,
- Glass
- Wood
- Plastic and rubber

Properties

- is transparent, or clear,
- are lightweight and flexible with low melting points.
- is lightweight and does not conduct heat easily.
- is brittle but has a very high melting point, allowing it to
- are strong, have a high melting point and conduct heat and electricity easily.

Functions and examples of use

- This makes it useful for holding hot objects without getting burnt.
Example – test tube holder
- They are used to support hot objects without melting.
Example – gauze mat
- They are useful in cool temperature situations where flexibility is needed.
Example – Bunsen burner hose
- allowing the contents to be clearly seen.
Example – beaker
- withstand very high temperatures without breaking.
Example – crucible

a Metals are _____

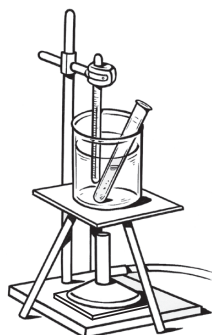
b Porcelain, a type of ceramic, _____

c Glass _____

d Wood _____

e Plastic and rubber _____

- 3 Label the type of material each piece of equipment is made from for the apparatus shown in the diagram.



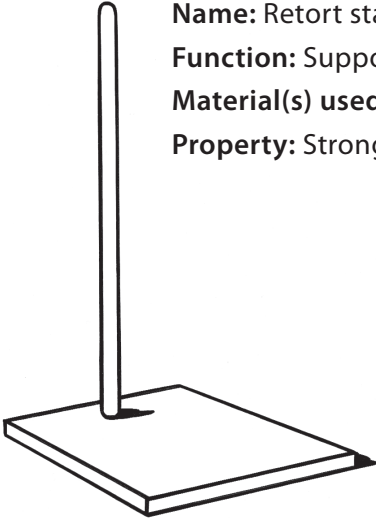
Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Most pieces of laboratory equipment are designed for specific functions. The design of an object refers to the way it is shaped or arranged. The function of a piece of equipment means what it is used for.

- 1 The information in the table is mixed up. Write the letters and number from each of the three columns that best match each piece of equipment. The first one has been done for you.
- a Crucible F 4 b
- b Measuring cylinder
- c Beaker
- d Evaporating basin
- e Conical flask
- f Test tube

FUNCTION	DESIGN	HOW THE DESIGN IS RELATED TO THE FUNCTION
A Measuring out volumes of liquids	1 Short and cylindrical with a large base	a Liquids can be swirled without spilling
B Crystallising salts by evaporating water from a solution	2 Tall and thin with markings evenly spaced along the side	b Lots of heat can be contained
C Mixing liquids by swirling	3 Small and thin	c Water can be evaporated quickly
D Holding large volumes of liquids	4 Small with a lid	d Chemicals are not wasted
E Holding small amounts of liquids or solids	5 Triangular with a large base but a narrow opening	e Small volumes can be accurately measured
F Strongly heating small amounts of solid	6 Bowl-shaped with a large opening	f Contents are easily heated

- 2 Complete the information for each piece of laboratory equipment on the next page. The one below has been done as an example.



Name: Retort stand

Function: Supporting objects

Material(s) used: Metal

Property: Strong

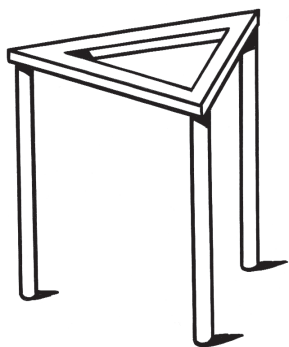
This property makes the material suitable for the function of this equipment because: The objects supported can be heavy and the stand won't bend or break.

Design: Large base

This design is suitable for the function of this equipment because: It makes the stand difficult to tip over.

A material's *properties* include its strength, colour, melting point and density.

a



Name: _____

Function: _____

Material(s) used: _____

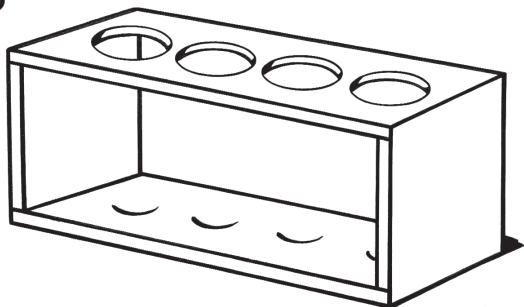
Property: _____

This property makes the material suitable for the function of this equipment because:

Design: _____

This design is suitable for the function of this equipment because:

b



Name: _____

Function: _____

Material(s) used: _____

Property: _____

This property makes the material suitable for the function of this equipment because:

Design: _____

This design is suitable for the function of this equipment because:

- 3 Ryan wanted to measure the change in temperature that occurred when a salt dissolved in water. None of the containers in the lab seemed to be suitable. His container needed to:
- be able to hold 100 mL of water
 - allow a thermometer to be used to measure the temperature change
 - allow him to stir the water to make sure the salt all dissolved and the water temperature was even
 - let very little heat be lost from the container.

Design a container that would be suitable for Ryan’s experiment.
Draw it in your workbook and label it with the properties and design features you have chosen to suit its function.

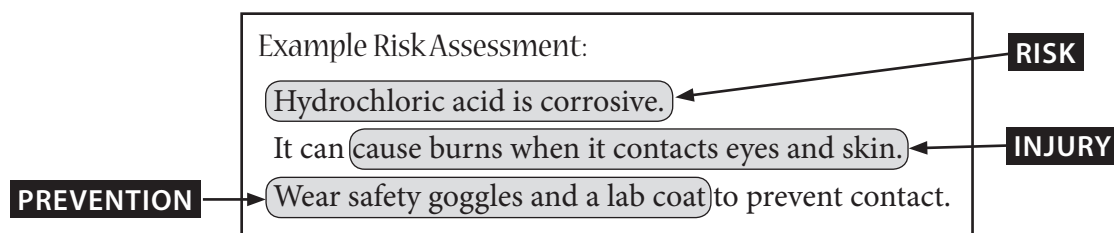


Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Safety is the most important factor in any experiment. Every experiment should be assessed for possible risks before it is conducted. No knowledge is worth gaining if someone hurts themselves to gain it.

All experiments have some risk. Even simple experiments could involve slipping on spilt liquid or cutting someone with a sharp object. Many experiments have more than one risk – all relevant risks should be mentioned in a risk assessment.

The *labels* of chemical bottles are a good source of information for risk assessments.



When assessing the risks involved in an experiment, remember RIP:

• **R is for Risk**

What is the object or chemical that is causing the risk? Describe how it is a risk.

• **I is for Injury**

What injuries could it cause? Describe them.

• **P is for Prevention**

What should be done to prevent the injuries occurring?

1 Label each statement as a *risk*, *injury* or *prevention* measure.

- a** Toxic ammonia gas _____
- b** Cuts to hands _____
- c** Clamp the thermometer securely using a retort stand, boss head and clamp. _____
- d** Irritation to lungs _____
- e** Sharp glass from broken beaker _____
- f** Allow all equipment to cool before packing up. _____

2 For each experiment described below, identify **one** risk (R), **one** injury (I) the risk could cause, and **one** way the potential injury could be prevented (P).

- a** Hannah heated 200 mL of water in a beaker using a Bunsen burner.

R: _____ I: _____

P: _____

- b** Chris and Kamal dropped golf balls from a balcony and timed how long they took to reach the ground.

R: _____ I: _____

P: _____

- c** Jin, Amber and Michelle dissected a chicken wing.

R: _____ I: _____

P: _____

3 Write **one** of the risk assessments from question 2 in sentence form.

Hint: Use the Example Risk Assessment above as a guide.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

The tests scientists use to gain information can be divided into two categories:

- destructive
- non-destructive.

Destructive testing

When a destructive test is used, the substance or object being tested is either destroyed or permanently changed so that further tests would not be valid. This means the results of future tests may be influenced by the previous test.

A valid experiment is a fair test.

Examples of destructive testing include:

- measuring how much weight it takes to break a piece of string, because the string is broken
- dissecting a chicken wing, because the parts cannot be joined back together.

1 Describe another example of destructive testing and explain why it is destructive.

Non-destructive testing

When a non-destructive test is used, the substance or object being tested is not permanently changed. Further tests will be valid because the object is still in its original form.

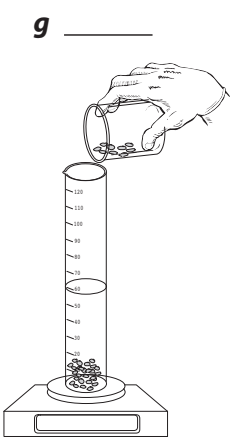
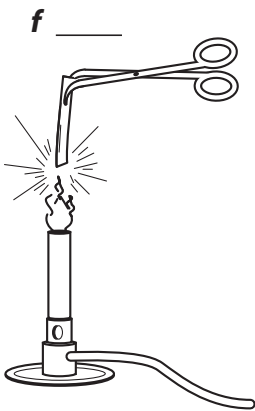
Examples of non-destructive testing include:

- using a thermometer to measure temperature
- using a pH probe to see whether a substance is acidic or basic.

2 Describe another example of non-destructive testing.

3 Identify whether the following tests would be destructive (D) or non-destructive (N-D).

- a Using liquid nitrogen to snap-freeze cells for examination under a microscope _____
- b Using X-rays to test an aeroplane wing for cracks _____
- c Testing the hardness of a porcelain tile by scratching it _____
- d Looking at a neon light with a spectroscope to determine its spectral fingerprint _____
- e Looking down a turbidity tube filled with creek water to measure how muddy the water is _____



4 Crash tests are used to test the safety of cars, but large structures such as buildings, bridges and dams are not usually tested in the same way to determine their safety.

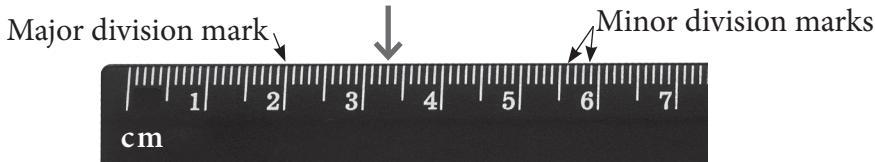
- a Propose one reason that this is the case. _____
- b Suggest one way the safety of buildings, bridges and dams could be tested. _____
- c Explain why it is important to test the safety of buildings, bridges and dams.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



Many scientific instruments now have digital displays, but scales are still found in many places. It is important that scales are read correctly. Drivers who don't read their speedometers correctly risk getting a fine!

1 Follow the steps below to work out the position of the grey arrow on the ruler.



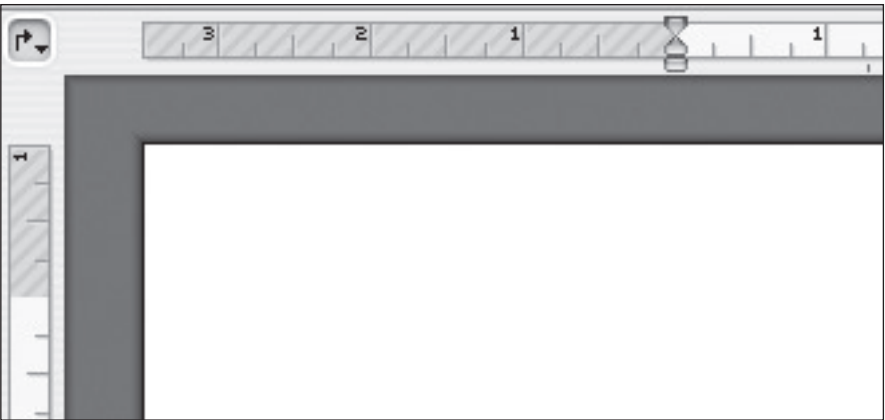
- Step 1:** Write down the major divisions on either side of the mark. _____ and _____
- Step 2:** Subtract the smaller major division from the larger major division. _____ - _____ = _____
- Step 3:** Count the number of minor divisions between the two major divisions and add 1.
_____ + 1 = _____
- Step 4:** Divide the answer in Step 2 by the answer in Step 3.
This is how much each minor division increases by. _____ ÷ _____ = _____
- Step 5:** Start from the smaller major division and add the amount calculated in Step 4 for each minor division until the mark is reached.
_____ + _____ + _____ + _____ = _____
- Step 6:** Write the unit of measurement after the answer in Step 5. Usually the unit of measurement is written on the instrument.

2 Place marks on the ruler at the top of this page at 4.7 cm and 2.0 cm.

3 Identify the reading shown on each instrument below.

a Measuring cylinder	b Thermometer	c Spring balance	d Voltmeter

4 Below is a screen grab of a Microsoft Word ruler, where the unit of measurement is centimetres.



- a** What is the left margin width as shown by the grey part of the ruler?

- b** What is the top margin width as shown by the grey part of the ruler?

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Ahmer wanted to know whether zinc or magnesium produced more heat when reacted with acid. He placed two test tubes in a test tube rack, then used a measuring cylinder to place 5 mL of acid in each test tube. He added zinc to one test tube and magnesium to the other test tube at the same time.

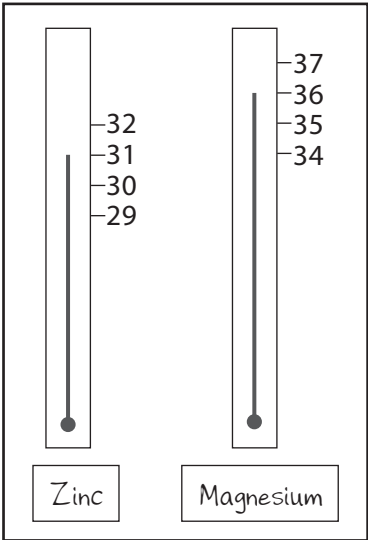
- 1 Draw a scientific diagram of Ahmer’s apparatus in the space at the right.
- 2 Explain one design feature of the measuring cylinder that makes it appropriate for measuring the acid.

- 3 Explain one property of the glass from which the measuring cylinder is made that makes it appropriate for measuring the acid.

- 4 Write a risk assessment for this experiment.

- 5 Would this test be considered destructive or non-destructive? Explain your answer.

Ahmer used thermometers to record the initial temperature of the acid and the highest temperature reached during each reaction for each test tube. The initial temperature of both test tubes was 23°C. The thermometers used to measure the highest temperature reached during each reaction are shown.



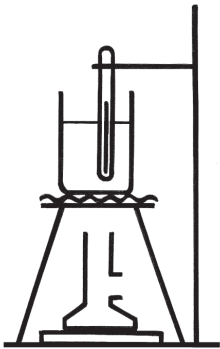
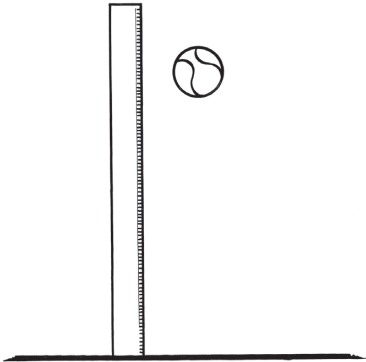
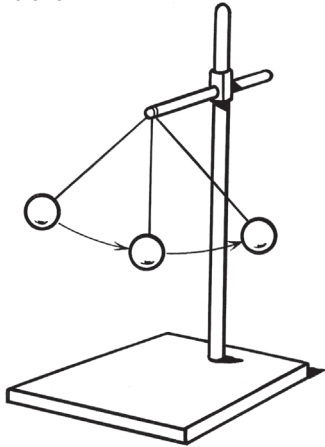
- 6 Write a conclusion for Ahmer’s experiment.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



Variables are the things about an experiment that can be different each time the experiment is performed. For example, in an experiment about how far a rugby ball could be kicked, some of the variables would be how hard it was kicked, what shoes were worn, whether it was windy and whether a pointy end or the rounder middle was kicked. Each of the things that can be different is a variable.

- 1
- Write down three variables for each experiment.
Hint: Consider how the experiment could be made different each time it was performed.

a Measuring the temperature of a beaker of water over a Bunsen burner 	b Measuring the bounce height of a ball 	c Timing the swing of a pendulum 

Types of variables

- The independent variable – *change*

In every experiment, one variable is deliberately changed. This is called the independent variable. Experiments should have only one independent variable.
- The dependent variable – *measure*

The variable that is measured to see the effect of changing the independent variable is called the dependent variable.

Experiments can have more than one dependent variable, but it makes the experiment more complicated. Usually it's better to concentrate on just one.
- Controlled variables – *same*

If more than one independent variable was used at a time, it wouldn't be clear which of them was having the effect seen on the dependent variable. So all variables other than the independent and dependent variables are kept the same. This makes the experiment a fair test. Variables that are kept the same are called controlled variables.

Example experiment

Rohan wanted to know if he could dissolve more salt in hot water than in cold water.

Independent variable:

He *changed* the temperature of the water.

Dependent variable:

He *measured* the mass of salt he could dissolve.

Controlled variables:

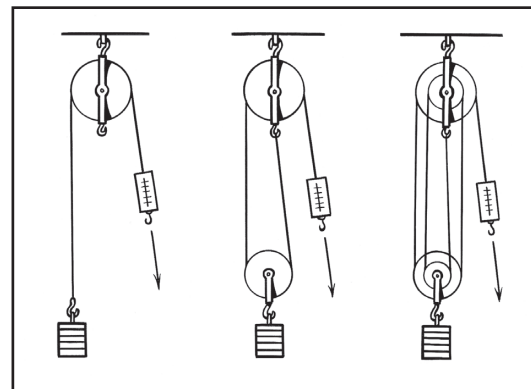
He kept these variables the *same*:

- the volume of water
- the amount of stirring
- the temperature of the room
- the balance used to measure the mass of the salt.

- 2 Read each experiment outlined below. Then identify the independent variable, the dependent variable and the controlled variables in each list.

- a** Rachel was doing an experiment in class on pulleys. She attached a 500 g mass to a single pulley. Then she attached the 500 g mass to two single pulleys and finally to two double pulleys. Each time, she tied a spring balance onto the end of a length of cotton string cut from the same roll and read the amount of force needed to lift the mass.

Rachel used the same spring balance and mass each time and made sure she pulled the spring balance slowly but steadily.



- ▲ Spring balance _____
- ▲ Mass _____
- ▲ Pulley system _____
- ▲ Cotton string _____
- ▲ Force _____
- ▲ Pulling the spring balance _____

- b** Alex told his friends he was better than any of them at playing the latest car race computer game. So Liam, Akshay, Kyle and Nishan challenged him to a fair test – whoever completed one lap of the same level in the quickest time would be the winner. Each played the game on Liam's hand-held game console at lunchtime in the playground, after they'd eaten their lunch.

- ▲ Game console _____
- ▲ Game level _____
- ▲ Player _____
- ▲ Number of laps _____
- ▲ Time since eating _____
- ▲ Time to complete a lap _____

- c** Taylor and Hayley bought superballs during a school excursion to the local science museum. They wondered which surface produced the highest bounce. They dropped one of the superballs from the top of a metre ruler onto concrete, wood, lino, tiles and carpet. Hayley held the ruler and dropped the ball and Taylor read the bounce height each time.

- ▲ Dropping the ball _____
- ▲ Superball _____
- ▲ Different surfaces _____
- ▲ Bounce height _____
- ▲ Method of measuring bounce height _____

- 3 Think about an experiment that could be used to test each hypothesis below. For each experiment, identify the independent variable, the dependent variable and at least three controlled variables.

- a Hypothesis:**
Taller people run faster.

Independent:

Dependent:

Controlled:

1 _____

2 _____

3 _____

- b Hypothesis:**
Longer strings can hold more weight.

Independent:

Dependent:

Controlled:

1 _____

2 _____

3 _____

- c Hypothesis:** *Magnesium metal cut into small pieces reacts faster with acid than large pieces of the same metal.*

Independent: _____

Dependent: _____

Controlled:

1 _____

2 _____

3 _____

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



An experimental control provides something to compare the results of experiments to. It shows that what is being tested really is causing the effect being seen. It is important to understand that an experimental control is different to controlled variables.

- 1 Use the word bank to complete the description of the experiment below.

Daniel read in his Science textbook that the bubbles in soft drink were made of _____. He thought he might find out how much carbon dioxide was in soft drink for his science project. He bought a 375 mL bottle of soft drink and asked his teacher if he could weigh it on the _____ at lunchtime. After writing down the initial mass, he opened the bottle and heard the sssssshhhh ... as the gas escaped. After ten minutes there were still _____ in the soft drink, so he left the bottle on the window sill overnight to allow enough time for the _____ to escape.

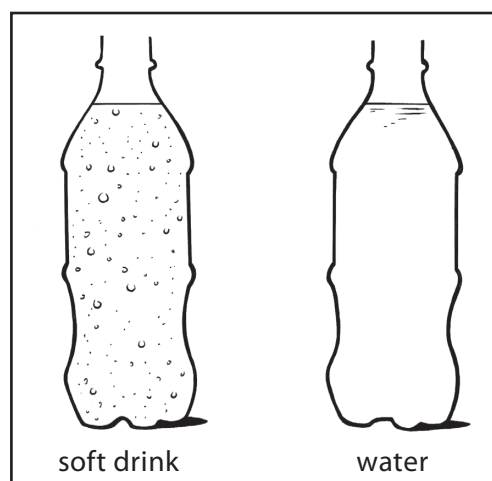
The next day, Daniel weighed the bottle again. He found that it had lost 3.50 g. Daniel concluded that there was _____ of carbon dioxide gas in a 375 mL bottle of soft drink. He also noticed that the level of _____ in the bottle seemed to have gone down. When he asked his teacher about it, she said that he may have lost some liquid because of _____ as well as losing the carbon dioxide gas. So Daniel repeated the experiment, but this time he also used a second 375 mL soft drink bottle filled with _____. He found the soft drink again lost 3.50 g and the bottle of water lost 0.39 g.

Daniel concluded that _____ of soft drink contains _____ of carbon dioxide.

Explanation: The bottle containing water in Daniel's experiment was an experimental _____. It provided something for him to _____ with the results of his soft drink experiment. This allowed him to validly determine the mass lost due to carbon dioxide escaping rather than evaporation.

WORD BANK

- liquid
- 3.11 g
- control
- evaporation
- 3.50 g
- electronic balance
- bubbles
- carbon dioxide
- water
- 375 mL
- gas
- compare



- 2 Describe an experimental control that could be used in each of the following experiments.

a Aim: To determine whether seeds need light to germinate.

b Aim: To measure whether heart rate is increased by exercise.

c Aim: To determine whether diet bars help people lose weight.

d Aim: To determine whether a new drug cures cancer.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Scientists use instruments to take measurements that help them describe and understand our world. However, taking measurements always involves errors. Errors are mistakes that occur when a measurement is made. They make measurements inaccurate and/or unreliable.

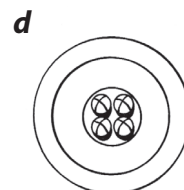
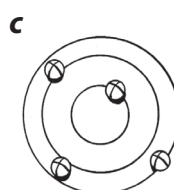
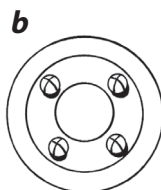
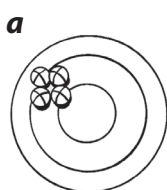
- **Accuracy** is described as how close a measurement is to the true value of what is being measured.
- A result is **reliable** if the same value is found every time a measurement is repeated.

Reliability is also known as precision.

A simple way to think about accuracy and reliability is to use a target analogy. If all the arrows hit the target close together, the shooter is reliable. If the average of the arrows is located on the centre of the target, the shooter is accurate.

- 1 Write each label below the appropriate target:

Accurate but unreliable,
Accurate and reliable,
Inaccurate and unreliable,
Inaccurate but reliable.



Measurement error

Errors can occur in a number of different ways. These ways include:

- incorrect calibration
- incorrect use of an instrument
- inappropriate choice of instrument
- uncontrollable or accidental errors.

Incorrect calibration

Calibrating an instrument involves checking whether it is measuring correctly. For example, if a 100 g mass is placed on an electronic balance, the balance can be expected to show exactly 100 g. If the balance reads 99 g or 101 g, it is not calibrated correctly.

- 2 Identify another scientific instrument that could be calibrated incorrectly. _____

Usually it is difficult or impossible to recalibrate a measuring instrument in the school lab, but any measurements made with an incorrectly calibrated instrument might be incorrect by the same amount each time. The person measuring might be able to adjust the measurements to take this into account.

- 3 Would a calibration error make a measurement:

a inaccurate? _____ **b** unreliable? _____

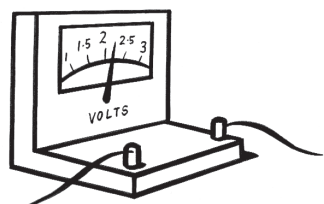
Incorrect use of an instrument

This error occurs when a person doesn't know how to use an instrument correctly. Common errors of this type include parallax error and not reading from the bottom of the meniscus in a measuring cylinder. Parallax error occurs when a measurement is not taken from directly in front of the scale. Often the same mistake is made in every measurement.

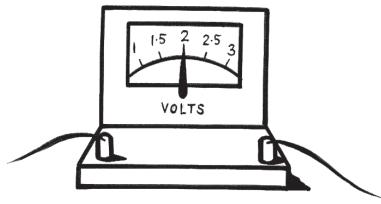
- 4 The same reading on a voltmeter is shown from three different angles.

a Write down the measurement that would be recorded in each case.

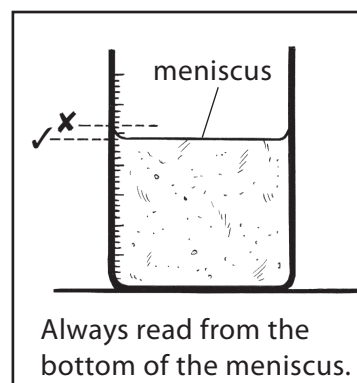
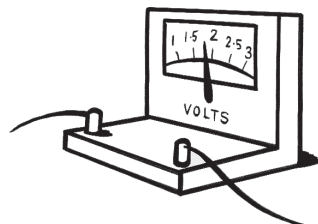
▲ _____ volts



▲ _____ volts



▲ _____ volts



b Tick the diagram with the measurement that would have no parallax error.

5 Would incorrect use of an instrument make a measurement:

a inaccurate? _____ **b** unreliable? _____

Inappropriate choice of instrument

Even with the proper use of correctly calibrated equipment, errors can be made if the instrument used is not appropriate. A measurement of the width of a hair or the length of a swimming pool would not be very close to the true value if it was measured using a 30 cm ruler!

6 Suggest an appropriate instrument to use to make the following measurements.

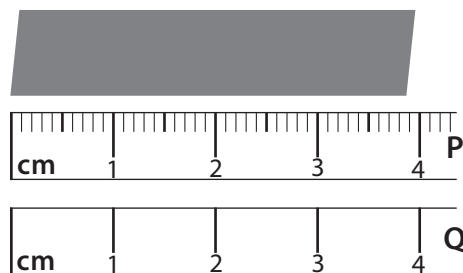
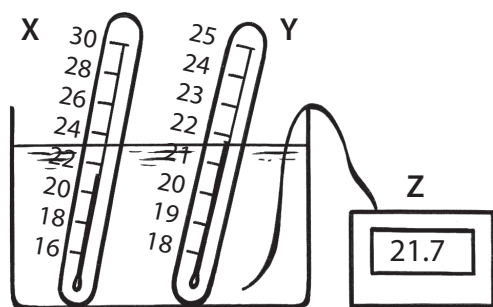
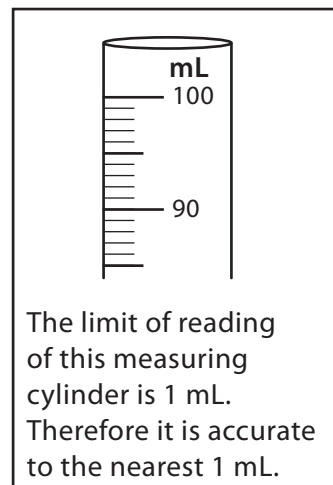
- a** The length of a swimming pool _____
b The mass of a person _____
c 5 g of sugar _____
d The time it takes a ball to fall 10 m _____

7 Would inappropriate choice of instrument make a measurement:

a inaccurate? _____ **b** unreliable? _____

If there is a choice of several appropriate instruments, the *limit* of reading of each one should be considered. The limit of reading is the smallest increment on an instrument's measuring scale. The most accurate instrument is the one with the smallest limit of reading.

8 Look at the instruments in the diagrams below.



a Circle the most accurate instrument in each diagram.

b State the limit of reading of each instrument. The first one is done as an example.

X – Accurate to the nearest 2°C Y – Accurate to the nearest _____ Z – Accurate to the nearest _____

P – _____ Q – _____

Uncontrollable or accidental errors

All the errors already mentioned usually affect only the accuracy of a measurement. Because the same error is being made over and over again, the measurement is usually reliable. Fair experiments try to keep all variables the same, except for the ones being deliberately changed or measured. However, errors still occur as it is very difficult to keep all variables completely controlled. These errors cannot be eliminated. Examples of uncontrollable or accidental errors are:

- small changes in the temperature of a room
- variation in the reaction time of a student using a stopwatch
- the natural unevenness of an object that is having its width measured
- the occurrence of accidents such as spilling or contaminating a sample.

9 Could uncontrollable or accidental errors make an experiment:

a inaccurate? _____ **b** unreliable? _____

10 Identify one other example of an uncontrollable or accidental error.



Good experiments are accurate, reliable and valid. **Accuracy** refers to how close measurements are to the true value. **Reliability** means the same results are achieved each time an experiment is repeated in exactly the same way. **Validity** is all about making sure an experiment is a fair test.

1 Predict a possible impact on society if scientists stopped ensuring that their experiments were valid.

Valid experiments

There are a number of criteria any experiment must meet to be considered valid.

- The experiment measures what it says it is measuring.
- Conclusions do not contain statements about things that haven’t been tested.
- All variables are kept controlled.
- An experimental control is used, if required.
- The experiment is accurate and reliable.

Measures what it says it is measuring

The experiment outlined below does not measure what it says it is measuring.

At her cousin’s wedding, Bethany noticed that most people who ate the brussels sprouts also wore glasses. She thought brussels sprouts must be bad for your eyesight. To test her idea, Bethany surveyed the people at the wedding to find out whether each person liked brussels sprouts or not, and whether or not they wore glasses. Her results are shown in the table.

	Likes brussels sprouts	Doesn’t like brussels sprouts
Wears glasses	32	4
Doesn’t wear glasses	6	17

Bethany concluded that eating brussels sprouts causes poor eyesight.

2 Do you think Bethany’s results support her hypothesis that eating brussels sprouts causes poor eyesight? Explain your answer.

Bethany’s conclusion could be correct. However, as people get older they lose taste buds and bitter-tasting foods such as brussels sprouts don’t taste quite so bad. Also, as people get older the lens in their eyes becomes less flexible and they often need to wear glasses. The people at the wedding, many of whom may be related, might have genes which mean they don’t develop eyesight problems when they are younger.

Bethany thinks she is measuring a relationship between poor eyesight and eating brussels sprouts, but she might be measuring a relationship between taste buds and age, and lens deterioration and age.

3 Suggest two ways that Bethany could improve her experiment to help her decide whether her conclusion is correct.

Conclusions do not contain statements about things that haven’t been tested.

4 Bethany’s sister Jasmine concluded from her sister’s results that people who don’t like brussels sprouts wear contact lenses. Is Jasmine’s conclusion valid? Explain your answer.

Keeps all variables controlled

Variables are the things in an experiment that can be different. If too many variables are different during an experiment, it is impossible to know which one is having the effect being measured.

In a fair experiment, one variable is deliberately changed – this is the independent variable. One variable is measured – the dependent variable. The dependent variable shows the effect of changing the independent variable. All other variables must be kept the same or they could also affect the dependent variable. If this happens, the scientist cannot identify which variable causes the effect seen on the dependent variable.

- 5** Jack wanted to find out whether a box with a glass lid heated up more than a box without a glass lid. He found two shoe boxes and recycled their cardboard lids. He punched a small hole in the side of each shoe box. Then he inserted a thermometer through the hole into each shoebox, and covered one shoe box with a sheet of glass. Jack put both shoe boxes in the sun. He measured the temperature of each shoe box every minute for 25 minutes.

a In Jack's experiment, what was:

▲ the independent variable? _____

▲ the dependent variable? _____

b Identify two variables Jack has to keep controlled to make his experiment valid.

- 6** Identify three variables that have to be kept controlled in experiments to test each of these hypotheses.

a Orange goldfish live longer than spotted goldfish. _____

b Cars can travel further when using premium fuel than when using regular fuel. _____

An experimental control is used, if required.

An experimental control provides something with which to compare experimental results. It can then be concluded whether the variable being tested really is having the effect being measured.

Ben wanted to know if adding fertiliser to the water used on tomato plants would make them grow more fruit. He planted a tomato seedling and watered it every day with liquid fertiliser. Four months later, the tomato plant had finished fruiting, and produced 34 tomatoes.

- 7** Can Ben write a valid conclusion from his experiment? Explain your answer. _____

- 8** How could Ben make his experiment valid? _____

Accurate and reliable

If an experiment is not accurate, it is not measuring the true value under investigation. Therefore it is not measuring what it says it is measuring, so it is also invalid.

- 9** Identify one way an experiment could be inaccurate. _____

If an experiment is unreliable, there must be a source of uncontrollable or accidental error. This means that all the variables that should have been controlled were not kept the same throughout the experiment. Therefore the experiment is also invalid.

Now check out the
PowerPoint slideshow
and the activity sheet
on the CD-ROM.



1 Rupinder wanted to know whether orange juice lost its Vitamin C more quickly in warmer temperatures. She made a dark purple indicator out of cornstarch and iodine. The higher the Vitamin C content of the juice, the lighter the colour of the indicator becomes. She squeezed three oranges and kept half of the juice in a light-proof, sealed container in the fridge, and half of the juice in an identical container on the bench.

Each day Rupinder used a measuring cylinder to measure 1.0 mL of indicator into two separate test tubes. She counted the number of drops of juice from the fridge needed to make the indicator in test tube 1 lose its colour. Using the second test tube, she counted the number of drops needed to make the indicator lose its colour using juice from the bench.

- a Identify the independent, dependent and three controlled variables in Rupinder’s experiment.
- ▲ Independent: _____
- ▲ Dependent: _____
- ▲ Controlled: _____
- b Rupinder also added drops of water to a third test tube containing indicator. Explain why she did this.
- _____
- c Identify the limit of reading of Rupinder’s measuring cylinder. _____
- d Rupinder used several different techniques to measure things in her experiment. Decide whether each technique listed below was accurate or inaccurate.
- ▲ Measuring the volume of the juice by counting the number of drops _____
- ▲ Changing the temperature of the juice by keeping it in the fridge or on the bench _____
- ▲ Judging when the colour was gone from the indicator by looking at it _____
- ▲ Using a measuring cylinder to put 1 mL of indicator in each test tube _____
- e Choose one technique in part d above that is inaccurate. Describe how its accuracy could be improved.
- _____
- _____
- f Overall, do you think Rupinder’s experiment could be considered accurate? Explain your answer.
- _____
- _____
- g Describe what Rupinder could do to judge whether her experiment is reliable.
- _____
- _____
- h Identify one feature of Rupinder’s experiment that makes it valid, other than features of its accuracy or reliability.
- _____

2 Outline a valid method to test the following hypothesis. *Hint:* Validity includes accuracy and reliability.

Cut flowers last longer when sugar is added to the water in the vase.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Scientists often use numbers to describe their observations. However, a number on its own doesn't have much meaning. If someone told you the number was 5.6, this wouldn't make any sense until they told you they were measuring the width of a room in metres, or the force needed to move a mass in Newtons. All measurements need information about what is being measured, a number and a unit.

The International System of Units

The International System of Units defines seven fundamental quantities. Each quantity has a symbol and a standard unit with its own symbol.

1 Use the word bank to complete the missing values in the table below.

QUANTITY	QUANTITY'S SYMBOL	STANDARD UNIT	STANDARD UNIT'S SYMBOL
Length	l	metre	
Mass	m		kg
Time		second	
Electric current	I	Amperes	
	T		K
Luminous intensity			cd
Amount of substance	n	mole	

WORD BANK

- l_v
- candela
- m
- temperature
- mol
- kilogram
- Kelvin
- A
- t
- s

2 But how do we know exactly how far a metre is, and how long is one second? Read the information on the history of systems of measurement, then answer the following questions.

The earliest known systems of measurement were developed sometime between the 3rd and 4th millennia BC in Mesopotamia, Egypt, the Indus Valley and possibly Persia. Their use spread over the centuries, and many different systems arose. Most of these traditional systems used parts of the body or objects from the environment as their units, so they were different from person to person and between different regions. The metric system was developed by a group of scientists in France, including the famous chemist Antoine Lavoisier. Based on the decimal system, the metric system was easier to use in calculations. Metric system units are defined so they can be consistently reproduced anywhere on Earth.

In 1963, the metric system was renamed the International System of Units. By 1971, the seven fundamental quantities in the table above were defined. By 2009, only three countries – Liberia, Myanmar and the United States of America – had not officially adopted the International System of Units.

- a Identify two civilisations that developed systems of measurement.

b Identify one disadvantage of traditional measurement systems.

c Identify two advantages of the International System of Units.

d **Research:** Find out how the metric system currently defines the standard unit of length.
- 

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Many of the quantities scientists use are found through calculations involving two or more fundamental quantities from the International System of Units. Some of these derived quantities are used so often they have been given their own symbols and standard units.

1 Draw lines to match each derived quantity with its symbol and units. Some have been done for you.

DERIVED QUANTITY	DERIVED QUANTITY'S SYMBOL	DERIVED UNIT	STANDARD UNIT	STANDARD UNIT'S SYMBOL
Frequency	p	s ⁻¹	Joule	Pa
Energy	f	kgm ² s ⁻²	Hertz	Hz
Pressure	E	kgm ⁻¹ s ⁻²	pascal	J

Deriving units

The formula used to calculate a derived quantity can also be used to derive its unit. This is very useful if you can't remember the unit for a derived quantity during an exam.

Distance is another word for length.

Example:

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

The unit of distance is metres, m.
The unit of time is seconds, s.

Therefore the unit of speed = $\frac{\text{the unit of distance}}{\text{the unit of time}} = \frac{\text{m}}{\text{s}} = \text{m/s} = \text{ms}^{-1}$

m/s is simply a more convenient way of writing $\frac{\text{m}}{\text{s}}$. Just as $\frac{1}{x} = x^{-1}$ in maths, any scientific unit with ⁻¹ after it means it is on the bottom of the fraction, so m/s = ms⁻¹.

When finding derived units, all the mathematical rules of fractions apply such as cancelling common factors and 'invert and multiply' if dividing with fractions.

2 Determine the derived unit for each of the following derived quantities. The first one has been started for you. Hint: To help you answer these questions, set your working out as shown in the example above.

Velocity and speed have the same unit.

a acceleration, given that acceleration is equal to velocity divided by time

the unit of _____ = $\frac{\text{the unit of velocity}}{\text{the unit of _____}} = \frac{\text{m/s}}{\text{_____}} = \frac{\text{m}}{\text{_____}} \times \frac{1}{\text{s}} = \frac{\text{m}}{\text{_____}^2} = \text{m/s} \text{_____} = \text{_____} \text{s}^{-2}$

b force, given that force = mass × acceleration

The unit of mass is kilograms.

c work, given that work is equal to force times distance

d kinetic energy, given that kinetic energy = ½ × mass × velocity squared

3 Prove that the unit of frequency of a wave is equal to the unit of velocity, ms⁻¹, divided by the unit of wavelength, m. Hint: Use information from Question 1.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Scientific notation makes very big and very small numbers easier to write. It is used when communicating measurements of the tiniest atomic particles to the vast distances of space. Numbers written in scientific notation have a number, b , multiplied by ten to the power of n , where n is a whole number; that is, $b \times 10^n$.

Writing large numbers in scientific notation

Example: Write 1 045 000 000 000 in scientific notation.

- Step 1** Write the first non-zero number: **1**
- Step 2** Put a decimal point and write all the following numbers. Ignore trailing zeroes unless they are significant: **1.045**
- Step 3** Count how many places the decimal point has moved from where it was in the original number: **12**
- Step 4** Write the number in Step 3 as the power of ten: **1.045×10^{12}**

Significant figures: Kate used an electronic balance to weigh out exactly 400 g of salt. Because Kate knows that the salt does not weigh 401 g or 398 g, the two zeroes are significant figures. In scientific notation, the mass would be written as 4.00×10^2 g.

- 1** Write the following numbers in scientific notation. Unless stated, trailing zeroes are not significant.

a 54 000 _____ **b** 67 000 000 000 _____

c 7800 (significant) _____ **d** 235 641 000 _____

Writing small numbers in scientific notation

Example: Write 0.0000000570 in scientific notation.

- Step 1** Write the first non-zero number: **5**
- Step 2** Put a decimal point and write all the following numbers. Always include trailing zeroes: **5.70**
- Step 3** Count how many places the decimal point has moved from where it was in the original number: **8**
- Step 4** Write the number in Step 3 as the power of ten, putting a negative before it: **5.70×10^{-8}**
- 2** Write the following numbers in scientific notation. Trailing zeroes are always significant.
- a** 0.000 45 _____ **b** 0.090 _____
- c** 0.000 000 000 03 _____ **d** 0.003 687 _____

Writing large scientific notation numbers as normal numbers

Example: Write 8.97×10^7 as a normal number.

- Step 1** Write the number without a decimal point: **897**
- Step 2** Calculate: power of ten (7) – how many numbers after the decimal point (2): $7 - 2 = 5$
- Step 3** Add on the number of zeroes you calculated in Step 2: **89 700 000**

Trailing zeroes come after the last non-zero number. Leading zeroes come before the first non-zero number.

- 3** Write these scientific notation numbers as normal numbers. Trailing zeroes are always significant.

a 3.4×10^4 _____ **b** 5×10^2 _____

c 54.9×10^3 _____ **d** 5.50×10^6 _____

Writing small scientific notation numbers as normal numbers

Example: Write 7.56×10^{-4} as a normal number.

- Step 1** Write the number without the decimal point, leaving space in front: **756**
- Step 2** Calculate: Power of ten, not including the negative sign (4) – how many numbers before the decimal point (1): $4 - 1 = 3$
- Step 3** Add on the number of zeroes you calculated in Step 2 to the front: **000 756**
- Step 4** Put a 0. in front: **0.000 756**

- 4** Write the following as normal numbers. Trailing zeroes are always significant.

a 3.4×10^{-4} _____ **b** 5×10^{-2} _____

c 54.9×10^{-3} _____ **d** 5.50×10^{-6} _____



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

The number of significant figures a measurement has indicates how accurate it is. Accuracy refers to how close a measurement is to the true value of what is being measured. For example, if an electronic balance is used to weigh 6.84 grams of copper sulfate, it would be incorrect to say there was 6.8 g or 6.843 g.

The 6, 8 and 4 in 6.84 g are all significant and give meaning to the measurement – they are significant figures. 6.84 g is more accurate than 6.8 g.

You will need a calculator to complete some of these activities.

Rules for significant figures

- **Leading zeroes** are not significant. If 4.3 g was written as 0.004 3 kg, the measurement still only has two significant figures.
- **Trailing zeroes** in decimal places are significant. 2.5 g could actually be 2.51 g or 2.48 g. 2.50 g shows it is certain that the second decimal place is 0.

Trailing zeroes come after the last non-zero number. Leading zeroes come before the first non-zero number.

Ambiguity means doubt or uncertainty.

Trailing zeroes before decimal places are ambiguous. For example, the trailing zeroes in 3 500 kg could be significant or could just be there to fill in places. It is best to use scientific notation to avoid this ambiguity. If the trailing zeroes are significant, 3.500×10^3 kg is written, which has four significant figures. If they are not significant, 3.5×10^3 kg is written, which has only 2 significant figures. If there is no further information, trailing zeroes are assumed to be significant.

1 Identify the number of significant figures in each of these numbers.

- | | | | |
|-----------------------|-----------------------|------------------------|-------------------------------------|
| a 8.97 _____ | b 0.05 _____ | c 4.00 _____ | d 4.2×10^{-4} _____ |
| e 29.010 _____ | f 54 031 _____ | g 0.013 0 _____ | h 42 000 _____ |

Rounding off

When rounding off significant figures:

- the numbers 0, 1, 2, 3 and 4 are rounded down
- the numbers 5, 6, 7, 8 and 9 are rounded up.

Example 1: Round off 0.001 249 843 33 to two significant figures.

Two significant figures means the number will either be 0.001 2 or 0.001 3, depending on whether it is rounded up or down. Look at the next significant figure, 4. Since it is a 4, round down to 0.001 2.

Example 2: Round off 5.687 to three significant figures.

Three significant figures means the number will be either 5.68 or 5.69, depending on whether it is rounded up or down. Look at the next significant figure, 7. Since it is a 7, round up to 5.69.

2 Round off the following numbers to the number of significant figures (sf) indicated.

- | | |
|---|---|
| a 0.000 011 489 (3 sf) _____ | b 6.324 81 (3 sf) _____ |
| c 0.023 (1 sf) _____ | d 54.466×10^{-5} (3 sf) _____ |
| e 560 (2 sf) _____ <i>Hint: Use scientific notation.</i> | |

Calculating with significant figures

Usually, each of the numbers used in a calculation has a different number of significant figures. The number of significant figures used in the answer depends on whether the numbers are multiplied/divided or added/subtracted.

When a calculation has many steps, don't round off until the final answer. Write down the unrounded final answer first in case you need the entire number for further calculations. Then write a rounded off answer.

Multiplying and dividing

The answer is rounded off to the smallest number of significant figures of the numbers used in the calculation.

Example: $5.786 \times 34.1 = 197.302 6 = 197$

5.786 has four sf. 34.1 has three sf. Therefore the answer is rounded off to three sf.

3 Perform the following calculations, rounding off each answer to the correct number of significant figures.

- a** 5.6×890 _____
b $0.002\ 8 \div 0.000\ 103$ _____
c $1.00 \div (6.5 \times 10^{-3})$ _____
d $16\ 500\ \text{g} \div (2.4 \times 10^3\ \text{kg})$ _____
e $23.46 \times 1.02 \div 5.47$ _____

Be sure to convert into the same units before performing calculations.

Different calculators enter scientific notation numbers differently. Try these ways or ask your science or maths teacher.

Example: 6.9×10^{-7}

1 Use the EXP button:

$6 \ . \ 9 \ \text{EXP} \ - \ 7$

2 Use the $\times 10^x$ button:

$6 \ . \ 9 \ \times 10^x \ - \ 7$

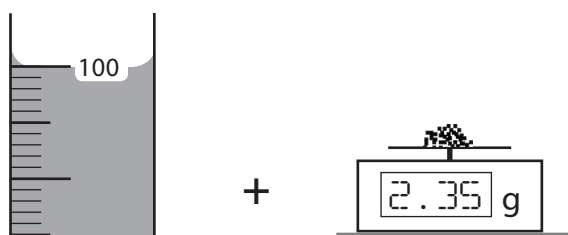
3 Use the second function button:

$6 \ . \ 9 \ \times \ 2\text{ndF} \ 10^x/\log \ (\ - \ 7 \)$

Adding and subtracting

The answer is rounded off to the smallest number of decimal places of the numbers used in the calculation.

Example: Calculate the mass of the salt water when the salt is added to the water.



1 mL of water has a mass of 1 g

$$\begin{array}{r} 100.??\ \text{g} \\ + \\ 2.35\ \text{g} \\ \hline 102.??\ \text{g} \end{array}$$

Explanation: Because the mass of the water is only accurate to the nearest 1 g, it is not known if its mass was really 100.3 g or 99.8 g, so the final mass of salt water can't be known any more accurately than 102 g. 100 g has no decimal places, 2.35 g has two. No decimal places is smaller than two, so no decimal places are included in the answer.

Numbers in scientific notation need to be written with just one number before the decimal place before they can be used in calculations.

Example: $675.89 \times 10^3 + 32.4 \times 10^2 = 6.758\ 9 \times 10^5 + 3.24 \times 10^3 = 679\ 130 = 6.79 \times 10^5$

4 Perform the following calculations, rounding off to the correct number of significant figures.

- a** $0.008 + 0.000\ 7$ _____
b $88.90 - 0.09$ _____
c $2\ 300 + 72.4$ _____
d $14.76 + 8$ _____
e $1.0\ \text{m} - 24.5\ \text{cm}$ _____
f $20.98 + 2$ _____
g $2.1 \times 10^{-4} + 3.3 \times 10^2$ _____
h $45.6 \times 10^{-12} - 8.97 \times 10^{-13}$ _____

5 Perform the following calculations, rounding off to the correct number of significant figures.

- a** $10.0\ \text{g} \div 2.5\ \text{kg}$ _____
b $65.753 + (49 \times 10^{-2}) - 8$ _____
c $5\ 600 \div 5.5 \times 10^{-9}$ _____
d $1.008 \div 6.02 \times 10^{23}$ _____
e $6.4 \times 10^{-1} - 0.32$ _____

Pure numbers such as the 2 in $2\pi r$ are not included when working out significant figures in calculations. However, special numbers with their own symbol, such as π or c for the speed of light, are included. For example, π is often rounded off to 3.14 and the speed of light to $3.0 \times 10^8\ \text{m/s}$.

6 a Calculate the circumference of a circle of radius 3.6 cm given that Circumference = $2\pi r$. _____

b Calculate the time in seconds it takes a radio wave to travel to a satellite if the distance is 40 000 km and the radio wave travels at the speed of light.

Hint: time = distance/speed. _____

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



Orders of magnitude are used to describe how big or small something is. For example, the order of magnitude of the width of a hair is very small, about 10^{-6} m or 0.000 001 m. However, its length is in the order of centimetres or 10^{-2} m. The distance to the next town would be in the order of kilometres, or 10^3 m.

Orders of magnitude are given a name and a symbol that can be used as a prefix in front of the unit of measurement. For example, 6.7 μ L is easier to write than 0.000 006 7 litres. The unit of measurement is litres, L, and the prefix is micro, μ .

1 Fill in the spaces in the table below.

PREFIX	SYMBOL	MULTIPLIES BY	ORDER OF MAGNITUDE	MEANING	SPECIAL TERMS
tera	T	$\times 1\,000\,000\,000\,000$	10^{12}	one trillion	
giga	G	$\times 1\,000\,000\,000$		one billion	
mega	M		10^6		
kilo			10^3	one thousand	1 tonne (mass)
deci	d	$\times 0.1$	10^{-1}	one tenth	
centi		$\times 0.01$		one hundredth	
milli		$\times 0.001$	10^{-3}		
micro	μ		10^{-6}	one millionth	
nano	n	$\times 0.000\,000\,001$			1 Angstrom, $\text{\AA} = 10\text{ nm}$ (length)
pico	p	$\times 0.000\,000\,000\,000\,1$	10^{-12}	one trillionth	

2 Write the equivalent measurements in the spaces in the table.

KILO		NO PREFIX		DECI		MILLI
54 kilometres	=	54 000 metres	=	540 000 decimetres	=	54 000 000 millimetres
	=		=		=	453.9 millilitres
	=		=	1.6 decivolts	=	
	=	0.2 amps	=		=	
0.001 kilograms	=		=		=	

3 Fill in the boxes to order the measurements from biggest (1) to smallest (5).

Hint: Convert each measurement to grams.

☐ 0.002 kilograms ☐ 20 000 grams ☐ 20 milligrams ☐ 2 000 centigrams ☐ 0.002 megagrams

4 Identify the order of magnitude of the following numbers. The first few are done as an example.

Hint: Write each number without a prefix for the unit, then write the number using scientific notation.

- a 3.5×10^{-9} 10^{-9}
- b 85 000 10^4
- c 70 Gm 10^{10}
- d 34 _____
- e 2.1×10^5 _____
- f 54 pm _____
- g 2.7 mm _____
- h 67 944 300 _____
- i 0.000 036 _____
- j 83×10^{-6} _____

The order of magnitude of familiar objects

5 Draw each object listed in the appropriate place on the scale below. The queen honeybee has been done as an example.

Notice that this is a special scale, called a logarithmic scale. It increases by multiples of 10. Each increment is ten times bigger than the previous one. Logarithmic scales are used when there is a very big difference in the size of each increment.

- | | | |
|---------------------------------|-------------------------------|-----------------------------|
| • A bacterium – 2 μm | • The Sun’s diameter – 1.4 Gm | • Mt Everest – 8 848 m |
| • A human – 1.7 m | • A virus – 100 nm | • A raindrop – 1.5 mm |
| • Earth’s diameter – 12.8 Mm | • A hydrogen atom – 100 pm | • The Eiffel Tower – 321 m |
| | | • A queen honeybee – 1.5 cm |

metres



Queen honeybee = 1.5 cm
= 0.015 m = 1.5×10^{-2} m

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Magnitude rummy (2–4 players)

The object of the game is to place all your cards on the table in sets of three or four.

Rules

- 1 Mix up the cards.
- 2 Each player is dealt one card in turn until all players have seven cards.
- 3 The remaining cards are placed facedown in the centre to form the deck. The top card is turned over and placed next to the deck to form a discard pile. The player to the left of the dealer goes first.
- 4 The player may either pick up the card from the discard pile or pick up the top card of the deck. Once they have seen their card, the player discards one card of their choice from their hand to the discard pile.
- 5 The next player to the left repeats step 4.
- 6 When a player makes a set of three or four, they may use their next turn to place the set face up on the table in front of them. They do not have the option of picking up or discarding a card on this turn. Only one set may be placed on the table at a time.
- 7 Other players may use their turn to add a matching card to a set of three on the table in front of any player. They do not have the option of picking up or discarding a card on this turn.
- 8 The game continues until one player has no cards remaining in their hand, and wins the game.

Magnitude match (whole class)

This game is best played on a sports field or other location with plenty of space and away from other classes. Adjust the deck to contain the same number of cards as there are players. Ensure all sets have three or four matching cards. Extra students could be in charge of handing out cards, judging winners or distributing prizes.

Rules

- 1 Mix up the cards.
- 2 Each player is given one card which they can't look at or show to anyone until the game starts.
- 3 All sets have three or four matching cards.
- 4 When the teacher blows the whistle or yells 'Go!', each player has to find the two or three other players with matching cards, until all cards in the set are located.
- 5 The first group to line up in front of the teacher and show them a complete set wins.

Magnitude snap (2 players)

- 1 All the cards are placed facedown on the table, evenly spaced.
- 2 Each player takes turns to turn over two cards.
If they are a match, the first player to call 'Snap!' removes the pair to their winning pile.
The player with the most cards when all are removed from the table wins.

Ideas

- Enlarge the card stencils onto A3 size for bigger cards.
- Laminate the cards so that they last longer.
- Back-to-back photocopy the cards with a pattern to prevent cards from being see-through. Create a page-sized box in Microsoft Word, then use 'Shape Fill' to fill it with a pattern.
- Each deck could be given a different pattern/colour/symbol for easy sorting.
- The rules could be photocopied, laminated and kept with each deck.

Magnitude rummy cards

2 terametres	2 Tm	2 000 000 000 000 m	2×10^{12} m
2 gigametres	2 Gm	2 000 000 000 m	2×10^9 m
2 megametres	2 Mm	2 000 000 m	2×10^6 m
2 kilometres	2 km	2 000 m	2×10^3 m
2 decimetres	2 dm	0.2m	2×10^{-1} m
2 centimetres	2 cm	0.02 m	2×10^{-2} m
2 millimetres	2 mm	0.002 m	2×10^{-3} m
2 micrometres	2 μ m	0.000 002 m	2×10^{-6} m
2 nanometres	2 nm	0.000 000 002 m	2×10^{-9} m
2 picometres	2 pm	0.000 000 000 002 m	2×10^{-12} m

- 1 Write these numbers in scientific notation.

a 65 444 _____

b 0.000 032 0 _____

c 450 nm _____
- 2 How many significant figures does each of the following numbers contain?

a 98 _____

b 1.2×10^{-6} _____

c 0.000 005 41 _____

d 9.60 kg _____
- 3 Round off these numbers to the stated number of significant figures (sf).

a 54.629×10^5 (3 sf) _____

b $9.893\,861\,1 \times 10^{-7}$ (2 sf) _____

c 3 500 (2 sf) _____
- 4 Perform the following calculations, rounding off to the appropriate number of significant figures.

a $65 + 8.7$ _____

b $(6.02 \times 10^{23}) \times 24.79$ _____

c $7.60 \times 10^2 \div (1.10 \times 10^{-3}) \times (8.5 \times 10^{-1})$ _____

d The distance swum at 3.0 m/s for 65 s if distance = speed $\times$ time _____
- 5 Draw lines to match each order of magnitude with its prefix and symbol.

kilo

mega

micro

milli

10^6

10^{-3}

10^3

10^{-6}

μ

M

k

m
- 6 Identify the order of magnitude of these numbers.

a 398 000 000 _____

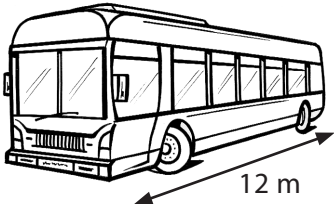
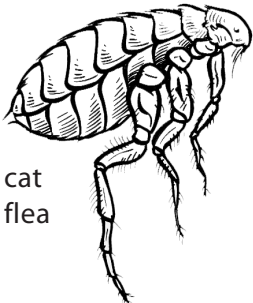
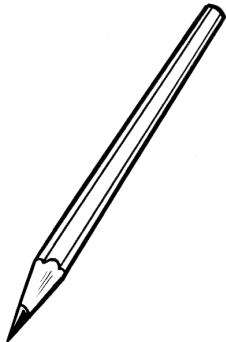
b 5.7×10^{-8} _____

c 0.009 _____

d 87 Mm _____

e 210×10^3 _____

f 5.4 Tm _____
- 7 Identify the order of magnitude of the length of these objects in metres.



- a _____

b _____

c _____

d _____

- 8 Complete the following table. The first one has been done for you.

NUMBER WITHOUT PREFIX	NUMBER WITH PREFIX	SCIENTIFIC NOTATION	NUMBER OF SIGNIFICANT FIGURES
9 007 m	9.007 km	9.007×10^3 m	4
	58.1 cm		
		1.2×10^{-5} g	

Well organised tables make it easier to record data and process it into graphs.

- 1 Unjumble the letters in brackets to complete the rules for drawing good tables.

Rule 1: Always draw (neils) _____ around and between the data.

Rule 2: Using pencil makes it easy to fix (ktemsius) _____.

Rule 3: Place all the data for one variable into the same (oumcIn) _____, not row.

Rule 4: Each column should have a heading describing the (aadt) _____ in the column.

Rule 5: Write the unit of measurement in the column (ieahdgn) _____, not with the data.

Rule 6: Adjust the data in a column so they all use the (mesa) _____ unit of measurement.

Rule 7: Put the independent (birlaeva) _____ in the first column and the dependent variables in the following columns.

Rule 8: Don't leave blanks. If the data is 0, write 0. If the data is (ont) _____ known, write –.

- 2 David conducted a survey to determine the most popular flavour of ice block for each student in his class. Put David's data into a table in the space at the right.

cola – 8, lemonade – 4, raspberry – 10,
lime – 3, orange – 5

In any experiment, there is one set of data where it is chosen which values will be tested – this is the **independent variable**. The data that is measured is the **dependent variable**.

	C	
R	O	W
	L	
	U	
	M	
	N	

Sometimes more than one thing is measured during an experiment or investigation. A separate column should be drawn for each thing being measured.

- 3 Charlotte performed an experiment to determine the properties of several elements such as electrical conductivity, colour, and malleability. Put Charlotte's data into a table in the space below.

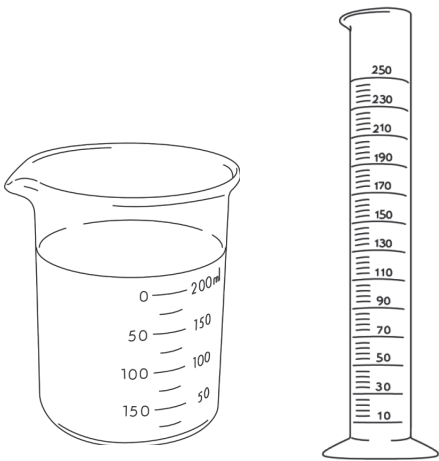
Charlotte's data
copper – conducts, orange, malleable; silicon – grey, conducts, brittle; sulfur – brittle, yellow, doesn't conduct; carbon – black, conducts, brittle; iron – not brittle, silver, conducts; aluminium – conducts, silver, malleable

If a material is *malleable*, it can bend without breaking. The opposite of malleable is brittle.

When using a table to compare things, the information is easier to read if the similarity or difference is in the same row.
When comparing, always include at least two similarities and two differences.

- 4 Complete the table below, comparing a measuring cylinder with a beaker.

Compare means identify similarities and differences between two or more things.



MEASURING CYLINDER	
Used to measure liquids to the nearest millilitre	Used to measure approximate amounts of liquids
	Found in science laboratories
Tall and thin	

Scientists often repeat experiments to test the reliability of their data. These repeat experiments are called trials. When recording the data from several trials, divide the column into sub-columns – one for each trial and a final one for the average. Only the average should be plotted if the data is used to draw a graph.

- 5 Gaya and Jason measured the time in seconds it took one drop of different liquids to slide down a wax paper ramp. They performed their experiment three times, obtaining the following results:
olive oil – 2.36, 2.25, 2.40; water – 0.45, 0.40, 0.50; honey – 20.09, 11.47, 13.33; ethanol – 0.12, 0.33, 0.27.
- a Complete Gaya and Jason’s table of results.

	TRIAL 1	TRIAL 2	TRIAL 3	AVERAGE

Viscous means resistant to flow.

- b Which liquid do you think is the most viscous?
Justify your answer.
- _____
- _____
- _____

HINT

To calculate an average:

1 Predict what the answer will be before doing the calculation. It should be a number somewhere in the middle of the data used to calculate the average.

2 Add up all the numbers.

3 If using a calculator, press =.

4 Divide the answer by how many numbers were added.

5 Compare the answer from Step 3 with the prediction from Step 1. If they are very different, check your calculation again.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



Timelines show the major milestones of a particular topic. They can show important achievements in history, or be used to plan future activities. Each milestone, achievement or activity is called an ‘event’. For example, a timeline could be constructed to show the events needed to complete an experiment, such as gathering information, planning, testing and writing up the report.

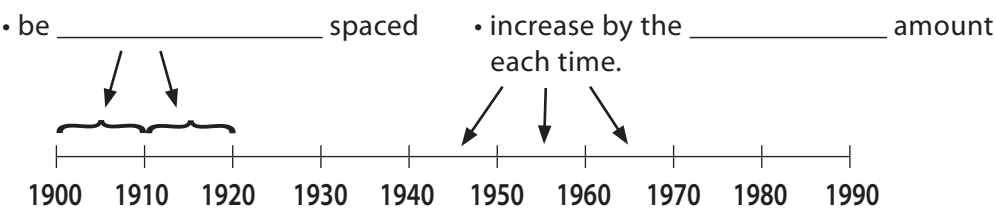
You will need a blank piece of A4 paper to complete question 3.

1 Use the word bank to complete the rules for drawing timelines.

WORD BANK

- names
- most
- line
- once
- brief
- equally
- diagrams
- pencil
- same
- large

Rule 1: When drawing a scale, the divisions must:



Rule 2: The timeline should take up _____ of the available space.

Rule 3: Using a _____ to draw a scale makes corrections easier.

Rule 4: Events on a timeline could include important information such as the _____ of the people involved, the place of the event and a very _____ summary of the event. More specific information about the exact time of the event could also be included. Sometimes it is useful to include _____.

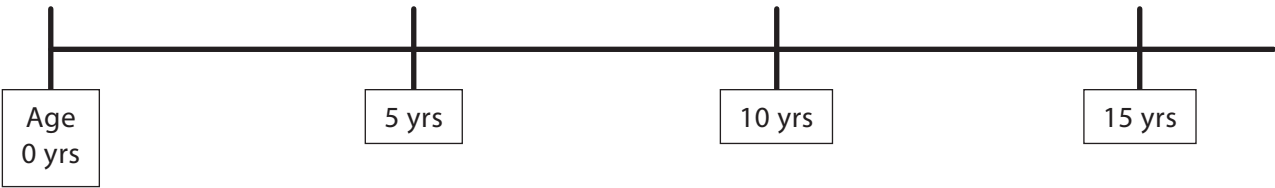
Rule 5: Each event should have a _____ joining the information to its date on the timeline.

Rule 6: If the events of a timeline have _____ gaps between them, the timeline may be broken using one of these symbols:



Timelines should only be broken _____.

2 Complete the timeline below with five events in your life.



- 3 Use the rules from question 1 and the information provided below to draw a timeline of the major achievements in our understanding of the structure of atoms. Use a blank piece of A4 paper.

For each event, include:

- the name of the scientist
- the date their model was proposed
- a brief summary of their model.

Hint: Read the entire article before you begin. Highlight or underline the information that you will include in the timeline. You may need to break the scale of the timeline.

Also include diagrams for at least five of the models of the atom.

The History of the Atom

The earliest references to the concept of atoms came from Indian philosophers of the 6th century BC. They theorised that atoms combined in pairs and trios to form complex objects. The word 'atom' comes from the Greek word *atomos* which means 'indivisible'. This was the name given by Democritus in the year 450 BC to describe what he believed was the smallest particle of matter. However, it was Aristotle's views in approximately 335 BC that dominated scientific thinking for the following 2000 years. He proposed that matter was composed of five elements – earth, air, fire, water and aether.

It was not until the science of Chemistry appeared in the 17th century that new progress was made. Chemists began to form theories based on observations from experiments rather than philosophy. In 1661, Robert Boyle argued against long-held beliefs by stating that matter was made of various combinations of corpuscles, or atoms. In 1789, the Frenchman Antoine Lavoisier defined the term 'element' – a substance that could not be further broken down by chemical methods.

In 1803, John Dalton, an Englishman, formulated the 'Law of Multiple Proportions' to describe how atoms join together in small, whole number ratios when they form chemical compounds. He proposed that all the atoms of one element are identical to one another, but different to the atoms of other elements. It was not until 1913 that Frederick Soddy predicted that the atoms of an element could exist as isotopes.

Experimentation with cathode rays in 1897 led J. J. Thomson to state that atoms were not indivisible particles as their name suggested, but were positively charged spheres with 'lumps' of negative charge he called 'electrons'. However, it was not until 1909 that a nuclear model of the atom was developed. Ernest Rutherford used the newly discovered tool of radioactivity to fire charged particles at a thin sheet of gold atoms. His results indicated that most of the atom was in fact empty space, with a small, dense positively charged nucleus and orbiting electrons. The positive particles in the nucleus became known as 'protons'. Niels Bohr refined Rutherford's theory in 1913 by suggesting that the orbit of each electron existed at a fixed energy level and that electrons could jump between orbits by absorbing or releasing specific amounts, or quanta, of energy.

Most physicists thought atoms behaved like particles until 1924 when Louis de Broglie proposed that atoms also behaved as waves. The mathematical explanation of this wave-particle duality was developed by Erwin Schrödinger in 1926 and was followed by Werner Heisenberg's Uncertainty Principle in 1927. The Uncertainty Principle states that while electrons have both a position and momentum, it is not possible to know both at the same time.

By this time, instruments had been developed to measure the exact mass of atoms. This allowed James Chadwick to explain in 1932 that isotopes are the result of different numbers of neutrons in the nucleus of atoms of the same element. In the 1950s, physicists used particle accelerators to further demolish the idea that atoms are indivisible. They showed that the protons and neutrons that make up atoms are themselves able to be divided into different quarks – called 'up', 'down', 'top', 'bottom', 'strange' and 'charm'.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



Models are used to help people understand how things work or how they are related to one another. They are often used for very small and very big things that are hard to visualise. In this activity, Microsoft PowerPoint is used to model how precipitation occurs when solutions of potassium iodide and lead nitrate are mixed together.

Step 1 Open PowerPoint by double-clicking on its icon.

Using PowerPoint 2003

- Click on 'Blank presentation' and click OK. Choose AutoLayout 1 for the first slide and click OK.

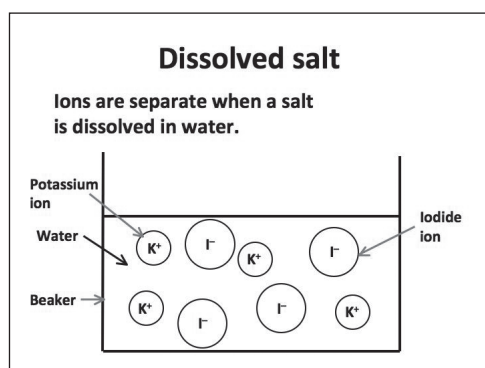
Using PowerPoint 2007

- PowerPoint will automatically create the first slide.

Step 2 Click to add the title "Precipitation of lead nitrate and potassium iodide". Leave customising the slides until the end.

Step 3 Click on 'New slide'. In PowerPoint 2003, choose AutoLayout 2. Click to add the title and the text shown in Slide 2.

Slide 2



To draw the beaker of dissolved ions:

Using PowerPoint 2003

- Make sure the Drawing Toolbar is visible at the bottom of the screen. If not, go to 'Tools', click 'Customise...' and in the 'Toolbars' tab check 'Drawing'. Click Close.
- Using the objects in the Drawing toolbar, draw the beaker of water. Make the beaker large, so it takes up most of the slide.
- Add one K^+ ion and one I^- ion. Text can be added to a circle shape by first drawing the circle, then clicking on the text box icon and clicking once on the circle.
- Colour each ion a different colour, using 'Shape Fill'. You may need to change the font colour and 'Shape Outline' to black.
- Superscript the + and - by highlighting them, then click on 'Format', then 'Font' and tick 'Superscript'. Click OK.
- Use the objects in the drawing toolbar to add the labels shown in slide 2.

- Copy and paste the K^+ and I^- ions until there are four of each in the beaker.

Using PowerPoint 2007

- Using the objects in the 'Home' tab, 'Drawing' section, draw the beaker of water. Make the beaker large, so it takes up most of the slide.
- Add one K^+ ion and one I^- ion. Text can be added to a circle shape by first drawing the circle, then clicking on the text box icon and clicking once on the circle.
- Colour each ion a different colour using 'Shape Fill'. You may need to change the font colour and 'Shape Outline' to black.
- Superscript the + and - by highlighting them, then click on the bottom-right corner of the 'Font' section and tick 'Superscript'. Click OK.
- Use the objects in the drawing toolbar to add the labels shown in slide 2.
- Copy and paste the K^+ and I^- ions until there are four of each in the beaker.

Step 4 Save the file by clicking on the disk icon. Name your file 'Precipitation'.

Don't forget to save your work regularly.

Always try to keep dot points short.

Step 5 Click 'New slide' to add the next slide. Click to add the text shown in Slide 3.

Slide 3

Precipitation

- Sometimes when two dissolved salts are mixed together, one type of positive ion will be attracted to one type of negative ion.
- Their attraction for each other is stronger than their attraction to the water.
- They become insoluble and form a precipitate.

Step 6 Click 'New slide' to add the next slide.

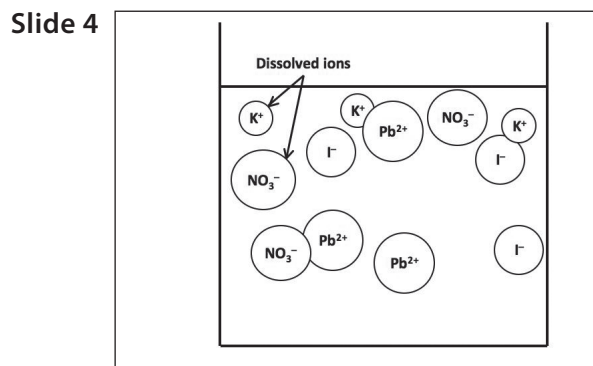
Using PowerPoint 2003

- Choose AutoLayout 12 – blank slide.

Using PowerPoint 2007

- In the 'Home' tab, 'Slides' section click on 'Layout' and choose 'Blank'.

Step 7 By copying and pasting the objects from Slide 2, draw a beaker containing three ions each of K^+ , I^- , Pb^{2+} and NO_3^- as shown in Slide 4. Place the ions towards the top of the beaker.



Step 8 Pb^{2+} and I^- form a precipitate, so they need to 'stick together' and fall to the bottom of the beaker. This can be done using Animations.

Using PowerPoint 2003

- Click on one of the Pb^{2+} ions. In the 'Slide Show' menu, click on 'Custom Animation'. A menu box will pop up.
- Click on 'Add Effect', then 'Motion Path' then 'Draw Custom Path' then 'Freeform'. Click once on a Pb^{2+} ion, then draw a path to an I^- ion and double-click to finish. Repeat this until all three Pb^{2+} ions and all three I^- ions are 'stuck together' as shown in slide 5.

Using PowerPoint 2007

- Click on one of the Pb^{2+} ions. In the 'Animations' tab, click on 'Custom Animation'. A menu box will pop up.
- Click on 'Add Effect', then 'Motion Path' then 'Draw Custom Path' then 'Freeform'. Click once on a Pb^{2+} ion, then draw a path to an I^- ion and double-click to finish. Repeat this until all three Pb^{2+} ions and all three I^- ions are 'stuck together' as shown in slide 5. Use the 'Play' button to preview your animation.

Step 9 Duplicate Slide 4 by:

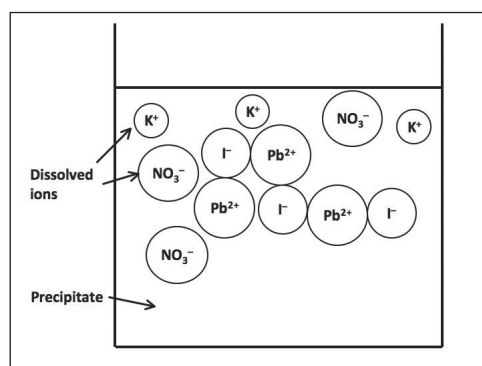
Using PowerPoint 2003

- clicking on 'Insert' then 'Duplicate Slide'.

Using PowerPoint 2007

- clicking on 'New slide' and choosing 'Duplicate Selected Slides'.

Slide 5



Step 10 On slide 5, group all the Pb^{2+} and I^- ions by holding down the 'Shift' key, and clicking on all the Pb^{2+} and I^- ions. Let go of the 'Shift' key then:

Using PowerPoint 2003

- Click on the 'Draw' menu in the Drawing toolbar at the bottom of the page and select 'Group'.

Using PowerPoint 2007

- In the 'Drawing Tools – Format' tab, go to the 'Arrange' section and click on 'Group'.

Step 11 Animate the ions so they fall to the bottom of the beaker by following the instructions in Step 8.

Step 12 Add a text box saying "Precipitate" and an arrow to the lead iodide. Animate it so that it appears after the ions have precipitated out by:

- Grouping the arrow and the text box as described in Step 10.
- Choosing 'Add Effect' then 'Entrance' then 'More Effects...', then 'Fade'. Click OK.

Step 13 Add a final 'Credits' slide and write in your name; adding any effects and animations you like. Click on the 'Slide Show' tab and choose 'From beginning' to see your slide show.

Step 14 Go back through your slides and customise them if you wish. For example:

Using PowerPoint 2003

- There are slide design options in the 'Format' menu, slide transition effects in the 'Slide show' menu and text effects in the 'Format' menu, 'Font' section.

Using PowerPoint 2007

- There are slide design options in the 'Design' tab, slide transition effects in the 'Animations' tab and text effects in the 'Home' tab, 'Font' section.

Step 15 Save and close the file.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



Models are used to explain observations and to make predictions. Often based on hypotheses or theories, models summarise what is known about a phenomenon, especially phenomena that are very large or very small. There are many different types of models. While each type of model is beneficial in its own way, they all also have limitations. A benefit is a way in which the model is helpful. A limitation prevents someone from understanding the model correctly or from getting useful information from the model.

1 Draw a line to match each type of model with its description.

TYPE OF MODEL	DESCRIPTION
Mathematical	a A three-dimensional physical model, proportionally smaller or larger, is constructed to represent the original.
Computer simulation	b An imaginary situation is used when physical experiments are difficult to conduct.
Diagrammatic	c Certain parameters are set and a computer generates a result.
Structural or scale model	d Equations are used to make predictions.
Thought experiment	e One thing is likened to something else; for example, an electric circuit is like water flowing through a pipe.
Descriptive	f A picture is drawn to show how various things are related.
Analogy	g Language is used to provide characteristics or features of a phenomenon.

2 The following text describes some of the benefits and limitations of using models. Highlight or underline the benefits in one colour and the limitations in another colour. Highlight or colour each box at the right with the corresponding colour to create a key.

Key:

☐	Benefit	☐	Limitation
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Models provide a better understanding of concepts that are difficult to comprehend or explain. Better understanding then leads to predictions that can be tested. The results of the test either support the model or produce modifications which improve the model.

In an effort to help people understand difficult concepts, models can become over-simplified or idealised, resulting in important information being overlooked or models that are actually incorrect. Models of the atom often show electrons following neat orbits close to a central nucleus, but this model has only a limited application to an actual atom. A model can be influenced by the interpretations of the person creating the model, so the model will be limited by their own understanding of the concept. Models described in one language may not be translated accurately into a different language.

It is usually cheaper, safer and faster to test a model than to test the real thing. For example, there are many benefits to testing the effects of atomic explosions using computer simulations rather than actually testing the explosions themselves. Models may allow for accurate predictions to be made, but they don't always help to explain the underlying causes; for example, when predicting the weather.

The information gained from models can make our lives safer, such as testing models of different building designs for their ability to withstand earthquakes. Scale models are cheap and easy to construct, but it is difficult to exactly replicate the materials and connections of a real structure. Potential drugs to cure diseases can be quickly identified using computer simulations to examine the complicated, microscopic three-dimensional structures of proteins. These drugs can be tested on particular species of animals chosen because they are easy to look after, are similar in biology to humans and breed quickly. This provides greater understanding of diseases and their potential treatments and cures. However, the assumptions that are made might make a model unrealistic. An animal's biology and therefore reaction might be subtly different from that of humans. Biologists run computer simulations on isolated populations and physicists assume friction is negligible.

Case study on the use of models – The search for new drugs to treat epilepsy

Epilepsy is an incurable disease affecting approximately 1 in 100 people. It is caused by abnormal electrical activity in the brain which produces sudden, uncontrollable misfiring of the nerve cells. As a result, the person suffers recurrent and unpredictable seizures.

3 Imagine you are a sufferer of epilepsy. Describe **one** impact seizures could have on your life.

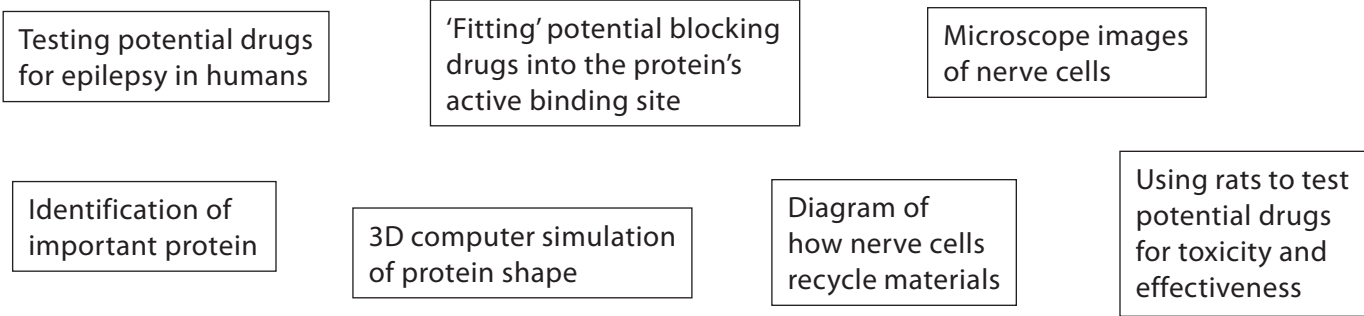
There are several different drugs available that can control epileptic seizures. They work by reducing nerve cell function. Unfortunately, one side effect of these drugs is that a person’s normal brain nerve cell activity is also reduced, which can lead to decreased motivation, difficulty concentrating and impaired mental and motor speed.

4 You are fortunate enough to be one of the approximately 75% of people whose seizures are controllable with these drugs. Describe **one** way your epilepsy could still impact on your life.

Researchers are trying to find a drug that can control seizures without any of the side effects. Starting with microscope images of nerve cells, they produced a diagram of how nerve cells replenish the materials they need to transmit messages to one another. From this diagram, they identified a protein that is important in this recycling of materials. By finding a drug that reduces the efficiency of this protein they might be able to prevent the misfiring of the nerve cells without any side effects.

Once the protein was identified, its genetic sequence was used in a computer simulation to predict its three-dimensional shape. The computer simulation then tried to ‘fit’ different drugs into the protein’s active binding site to disable it – like a key fitting into a lock. Drugs that fit well can be tested on rats with epilepsy to see whether they work. The best of the drugs that works in rats and is not toxic can then be trialled in humans.

5 Label the steps from 1 to 7 to show the order in which they were used to find potential new drugs for treating epilepsy.



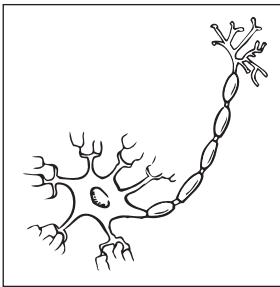
6 Identify the three types of models used by the researchers.

7 Identify one benefit the researchers gain from using models in their work.

8 Identify one limitation the researchers could experience from using models in their work.

9 Identify one benefit and one limitation of the diagrammatic model of the nerve cell shown.

BENEFIT	LIMITATION



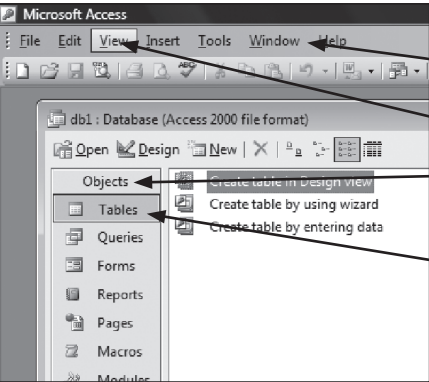
Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Databases make it easier to store and manage large amounts of data. They allow users to enter and extract only the data needed at a particular time, which makes the data more manageable for the user – that is, the person who uses the database. This activity uses Microsoft Access, but there are other database programs available such as FileMakerPro.

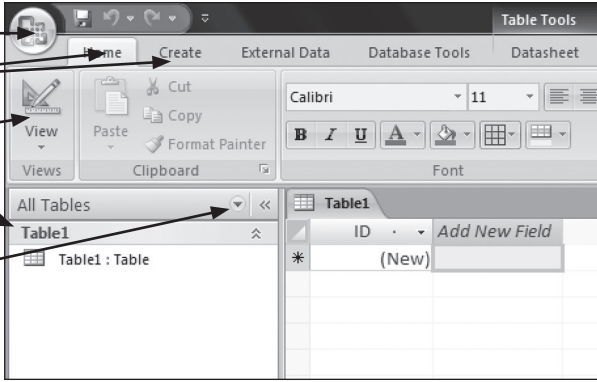
The database you make in this activity will store information about some of the elements of the Periodic Table. You will need a computer that has Microsoft Access 2003 or 2007 to complete this activity.

A database is made up of Objects, and each type of Object has a specific job to do. The main Objects are Tables, Queries, Forms and Reports. Objects are managed using the Objects List.

Access 2003



Access 2007



Office button
Tabs
Switch between views
Objects List
Click here to select different types of Objects. Double-click on an Object to open it.

As you follow the steps below to create a database, try working in pairs with one person reading the directions and one person working on the computer. Make sure you swap roles regularly.

1 Explain one advantage of working in pairs for this activity rather than individually.

Tables

The first step for any database is to create a table. Tables store data and can be customised to make sure users always enter the correct type of data. For example, they can be customised to prevent users accidentally entering their address when they should be entering their phone number.

Step 1 Open Microsoft Access by double-clicking on its icon.

Access 2003	Access 2007
Step 2 Select 'Blank Access Database' and click 'OK'.	Step 2 Select 'Blank Database' and type in 'Elements' as the file name.
Step 3 Type in 'Elements' as the file name, select where you want to save your database and click 'Create'. Double-click on 'Create Table by Entering Data'.	Step 3 Click on the small folder icon to the right of the file name to select a place to store your database. Click 'Create'. Access will automatically create a blank table when it creates the database.

The table you create will be quite small, but you can imagine that the table for a banking company that stores the names, addresses, account balances and all the other information for all their customers would be huge!

2 Identify one other place where you have seen a database being used.

Fields

Each column in a database is called a field. Like a column heading, it describes the type of information that will be stored in that particular field. For example, a bank database might have the fields *Name*, *Address*, *Account balance* and so on.

3 Identify at least four fields you would create to store information about different species of organisms living in an ecosystem.

Records

Each row in the table is called a record. In a bank database, one record would hold all the information for one customer, such as Harry Smith; 32 Moneybags Lane, Richmond; \$254.98.

The Primary Key

Access uses a Primary Key to help it keep track of all its records. This is a unique number identifying each individual record. You can choose a Primary Key, or allow Access to do it for you. Since the atomic number of each element is unique, it can be used as the Primary Key.

The atomic number of each element is unique. It is equal to the number of protons in the nucleus.

Step 4 Click on the disk icon to save the table and name it 'Elements'.

Access 2003 Step 5 Access will tell you there is no Primary Key defined. Click 'Yes' to create one. Double-click on the column name 'ID' and rename it "Atomic Number". Press Enter. Step 6 You will need eight fields in total. Click on 'Field1' then go to 'Insert' then 'Column'. Repeat this until there are eight fields. Rename the fields until the table looks like this:	Access 2007 Step 5 Double-click on the field name 'ID' and change the name to "Atomic Number". Press Enter. Step 6 Create another field by double-clicking on the field heading 'Add new field'. Call this second field "Element Name". Press Enter and Access will automatically be ready to enter the next field. Continue to rename fields until the table looks like this:
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ATOMIC NUMBER	ELEMENT NAME	SYMBOL	ATOMIC WEIGHT	METALLIC CHARACTER	MELTING POINT (°C)	BOILING POINT (°C)	USES
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4 Identify two other fields that could be included.

Step 7 Widen each field so the whole field name can be seen. Do this by moving the mouse arrow over the border between one field and the next, then click and hold the left mouse button and drag it wider.

Step 8 Save the table.

Step 9 Choose 'Design View' in the top left-hand corner. This allows properties to be set for the data that can be entered into each field. Setting the properties of a field stops users from entering the wrong type of data.

Step 10 Change the 'Data Type' of "Atomic number" from 'AutoNumber' to 'Number'. This allows you to enter the Primary Key, rather than have Access generate numbers automatically.

- Step 11** Properties can be set by clicking the text box containing the property to be changed. A drop-down arrow will appear on the right side from which selections can be made.
- As the only choices for Metallic Character are metal, semi-metal or non-metal, a drop-down box called a Combo box can be created. Click on 'Metallic character' then click on the 'Lookup' tab. Change 'Display Control' from 'Text box' to 'Combo box'. Change 'Row Source Type' from 'Table/Query' to 'Value List'. Now type the values into the 'Row Source' box exactly like this: "metal;semi-metal;non-metal".
 - Since no element has a symbol longer than 3 letters, click on the 'Symbol' field, then in the 'General' tab underneath, change 'Field Size' from "255" to "3". 'Field Size' determines how many characters can be typed into the field.
 - Change the 'Data Type' of the "Melting point" and "Boiling point" fields from 'Text' to 'Number' to ensure users only enter numbers.
 - Data on melting point and boiling point may have several decimal places. To enter decimal places into these fields, click on the 'Data Type' of "Melting point", change the 'Field Size' to "Single", change the 'Format' to "Fixed" and set the 'Decimal places' to "3". Repeat this for the "Boiling point" field.

Step 12 Save, then close the table by clicking on the cross in the top right corner of the table window.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

In databases, information is stored within tables but forms are used to put the information into the tables. To complete this activity, you need to access the database file you created in the previous activity ‘Using a database – designing’ and the Internet to collect information to enter into the form.

Forms

Form Objects are used to enter the data for each record. Forms should be designed to make entering data easy, especially when users might not understand how databases work and when lots of different users will use the database. Using a form to enter data means fewer mistakes are made. This is because forms:

Wizards make it easy to create Objects, but if you want to customise them, you need to use Design View.

- have security options for each field, so you can choose which data in a record users can change
- can display selected fields, which makes it faster and easier to enter data
- allow users to enter data into several different tables all at once, if the database has more than one table.

In this activity, you will create a form to enter information into the table you created in the previous activity. You will need to open the ‘Elements’ database you created in the previous activity. Find where you saved it then double-click on it to open it.

Access 2003

Step 1 In the Objects List window, click on ‘Forms’. Double-click on ‘Create form by using wizard’. Users need to enter data for all fields in the table, so click on the >> arrows to select all fields. Click ‘Next’. Choose the ‘Columnar’ layout and click ‘Next’. Choose whichever style you like best and click ‘Next’. “Elements Form” is a good name for the form; type it in and click ‘Finish’. The form will be added to the Objects List.

Access 2007

Step 1 Click on the “Elements” table you created in the previous activity. Click on the ‘Create’ tab, and choose ‘Form’. Access will automatically generate a form. Each field in the table has been given a heading and a field box for entering data into. Save the form; call it “Elements Form” then click ‘OK’. The form will be added to the Objects List.

- Step 2** Save, then close the Form.
- Step 3** Double-click on the newly created “Elements Form” to open it, then use it to enter data. The symbol >* adds a new record. < and > allow you to scroll through records and << and >> take you to the first and last records respectively.

Use the Internet to find information on the elements iron, silicon, gold, oxygen, hydrogen, aluminium, mercury, neon, carbon, zinc and lead. You will need to add the Atomic number and Element name before creating each new record. Try to find at least three uses for each element, separating each use with a comma. Good sites for information include www.webelements.com, www.chemicool.com and, if you have headphones, www.periodicvideos.com.

When an Object is created or changed, you need to save it. But when data is entered into a record or changed, it saves automatically.

- 1** Scientists sometimes measure temperature in Kelvin, K. Convert between degrees Celsius and Kelvin to complete the table at the right.
- Hint: 0°C = 273K

TEMPERATURE (°C)	TEMPERATURE (K)
25	
	183
	1 340
-186	

- 2** Define *absolute zero*.
Hint: Type ‘absolute zero’ into a search engine such as Google.
-
-
-

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

The real power of a database is its ability to search through data to extract specific information. Searching is done using Queries, and the results of searches are presented using Reports. You need to access the database file created in the previous activities ‘Designing’ and ‘Entering data’ to complete this activity.

Queries

Query Objects allow specific data to be found from a database, including across several tables at once if the database has multiple tables. Queries can also perform calculations. You are going to design a query to find out the uses of the metal elements.

Step 1 Open the “Elements” database you created in previous activities.

Access 2003 Step 2 In the Objects list window, click on ‘Queries’. Double-click on ‘Create query in Design View’.	Access 2007 Step 2 In the ‘Create’ tab, click on ‘Query Design’. In the ‘Show Table’ pop-up, click ‘Add’, then close the pop-up box.
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- Step 3** In the first column, choose ‘Element Name’ from the drop-down box. In the second column, choose ‘Symbol’, in the third column choose ‘Metallic Character’ and in the fourth column choose ‘Uses’.
- Step 4** In the ‘Element Name’ column under ‘Sort’, click ‘descending’. This will alphabetise the metal elements.
- Step 5** In the ‘Metallic Character’ column under ‘Criteria’, type =”Metal”. Include the quotation marks. This will ensure only data on metals is extracted from the database. It is not necessary to show that they are metals, so uncheck the check box ‘Show’.
- Step 6** Save your query and call it “Metal Uses”. Click on ‘Run’ to check your query results then close the query. If the query is wrong or produces an error message, go back through the previous steps.

Reports

Report Objects present the data found by a query. They should be designed so they are easy to read and look appealing. You are going to create a report on the “Metal Uses” query.

Access 2003 Step 7 Click on ‘Reports’ in the Objects List window. Double-click on ‘Create report by using wizard’. Step 8 In the Tables/Queries drop-down menu, choose ‘Query: Metal Uses’. Click on >> to select all fields. Click ‘Next’. Don’t add any grouping levels. Click ‘Next’. The sort order was set when we designed our query, so just click ‘Next’. Choose any layout options you want, then click ‘Next’. Choose any style options you want, then click ‘Next’. Step 9 Call your report “[Your Name] Metal Uses” then click ‘Finish’. Click on the Print icon to print your report.	Access 2007 Step 7 Click on the “Metal Uses” query in the Objects List, then in the ‘Create’ tab, click ‘Report’. Step 8 Click on ‘View’ and choose ‘Design View’. Customise your report using the options in the ‘Design’ tab. Step 9 Save your report and call it “[Your Name] Metal Uses”. Go to the Office button and print it.
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- Step 10** Close the report.
- Step 11** Now create a report on all the information in your database by following steps 7–9 for “Table:Elements” instead of “Query:Metal Uses”.
- Step 12** Close the database.

1 Identify which of the following tasks could be best accomplished using a database.____

A – Creating a graph from some experimental results	B – Typing up an assignment	C – Storing the results of a large survey	D – Creating a presentation for display on a data projector
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2 Identify a suitable computer program for accomplishing the other three tasks listed in Question 1.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

1 Use the word bank to complete the table by writing each term beside the correct description.

WORD BANK	• database	• query	• table	• field
• report	• form	• record	• wizard	• objects
• primary key	• data	• design view		

TERM	DESCRIPTION
a	Used to enter data into a database
b	View where objects can be customised
c	The data in a database for one person or element; a 'row' in a table
d	A step-by-step process for quickly creating Objects
e	Examples include tables, forms, queries and reports
f	Allows specific data to be extracted from the database
g	Where all the same data for each record is stored; a 'column' in a table
h	A unique number assigned to each individual record
i	A well presented way of viewing selected data from a database
j	Information collected from an experiment, survey or other source
k	A file consisting of Objects where data can be entered, stored, queried and reported on
l	Consists of fields and records; stores the data in the database

2 Mark each statement as true or false.

- a The ability to add security features is one advantage of using forms. _____
- b Databases are used when there is a small amount of data. _____
- c The main purpose of a database is to store, manage and retrieve data. _____
- d It is best to use a text box when there are limited choices for the data in a field. _____

3 a Outline one advantage of data being automatically saved by Access. _____

b Outline one disadvantage of data being automatically saved by Access. _____

4 Identify three fields you would use when creating a database containing contact information for your friends.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

1 – Drawing scientific diagrams

- Rule 1:** pencil. **Rule 2:** ruler. **Rule 3:** high. **Rule 4:** two. **Rule 5:** labelled. **Rule 6:** lines. **Rule 7:** touching. **Rule 8:** lid. **Rule 9:** Bunsen, beaker.
- a** measuring cylinder **b** gauze mat **c** heatproof mat **d** conical flask **e** filter funnel with filter paper **f** Bunsen burner **g** evaporating basin **h** test tube in a test tube rack **i** retort stand with boss head and clamp **j** tripod **k** beaker **l** thermometer
- Correct:** Pencil used, straight lines drawn using a ruler, No floating equipment, Equipment is in correct proportion to each other, Label lines touch equipment, Openings are not closed off. **Incorrect:** All equipment correctly labelled, About 10 cm high, Diagram is 2D, with no shading, No arrowheads on label lines

2 – Choosing lab equipment A

- Examples could include boiling point, hardness, electrical and heat conductivity, magnetism, lustre, state at room temperature.
- a** Metals are strong, have a high melting point and conduct heat and electricity easily. They are used to support hot objects without melting. An example is a gauze mat. **b** Porcelain, a type of ceramic, is brittle but has a very high melting point, allowing it to withstand very high temperatures without breaking. An example is a crucible. **c** Glass is transparent, or clear, allowing the contents to be clearly seen. An example is a beaker. **d** Wood is lightweight and does not conduct heat easily. This makes it useful for holding hot objects without getting burnt. An example is a test tube holder. **e** Plastic and rubber are lightweight and flexible with low melting points. They are useful in cool temperature situations where flexibility is needed. An example is a Bunsen burner hose.
- Retort stand, boss head, clamp, tripod, Bunsen burner, gauze mat – metal. Thermometer, beaker, test tube – glass. Bunsen burner tube – rubber. Heatproof mat – fibreglass. Thermometer liquid – alcohol

3 – Choosing lab equipment B

- a** F 4 **b** A 2 **e** C D 1 **f** D B 6 **c** E C 5 **a** F E 3 **d**
- a** Name: Tripod. Function: Supporting containers over a Bunsen burner. Material(s) used: Metal, steel. Property: For example, high melting point. This property makes the material suitable for the function of this equipment because Bunsen burner flames can reach over 1000°C so it won't melt. Design: For example, three legs. This design is suitable for the function of this equipment because it is stable and won't tip over.
b Name: Test tube rack. Function: Holding test tubes. Material(s) used: Wood. Property: For example, lightweight. This property makes the material suitable for the function of this equipment because test tubes are not heavy and the rack can be easily moved around. Design: For example, holes in the top piece of wood. This design is suitable for the function of this equipment because test tubes are easily placed in the rack and can't fall over and spill.
- Features of container could include: approximately beaker-sized, made of polystyrene/insulated, lid, holes for thermometer and stirrer in lid.

4 – Assessing risk

- a** risk **b** injury **c** prevention **d** injury **e** risk **f** prevention
- a** R – hot water, hot Bunsen burner or tripod, broken glass of beaker. I – burns to skin and eyes, cuts. P – wear safety goggles and lab coat, ensure equipment is set up in a stable way and away from the edge of the bench, wait until equipment cools before packing up
b R – falling objects (balls, people). I – concussion, head injuries, broken bones and bruising. P – rope off the area so people can't access it, ensure the way is clear before dropping the ball, stay away from the edge and do not lean over too far.
c R – sharp scalpel, infection. I – cuts, disease (e.g. bird flu, food poisoning). P – apply only gentle pressure to scalpel, wear gloves and safety goggles, wash and disinfect equipment and hands after contact.

- Example 2a:** Hot equipment and liquids can cause burns to the skin and eyes. Wear safety goggles and a lab coat and ensure all equipment is cool before packing it away.

5 – Destructive and non-destructive testing

- Examples may include: putting an object under stress to see how strong it is or cutting an object open because the object is destroyed; staining cells or using acid/base indicators because the substances being tested are chemically changed.
- Examples may include: weighing, measuring length, ultrasound.
- a** D **b** N-D **c** D **d** N-D **e** N-D **f** D **g** N-D
- a** Too expensive, too time-consuming **b** Build and test models, computer simulations **c** If the building/bridge/dam is not safe, it could collapse and kill people.

6 – Reading scales

- Step 1: 3 and 4. Step 2: $4 - 3 = 1$. Step 3: $9 + 1 = 10$. Step 4: $1 \div 10 = 0.1$. Step 5: $3 + 0.1 + 0.1 + 0.1 = 3.3$ cm
- a** 46 mL **b** 3.5 N **c** -0.1°C **d** 4 V
- a** 3.5 cm **b** 1 cm

7 – Quick quiz A

- For example, the measuring cylinder is tall and thin, so the volume of liquid can be measured accurately.
- For example, the glass is transparent, so the volume of liquid is easily seen.
- For example: Acid is corrosive. To protect from burns to the eyes and skin, safety goggles and a lab coat should be worn.
- Destructive, because the metals are dissolved in the test.
- Magnesium produces more heat than zinc when placed in acid.

8 – Experimental variables

- a** Amount of water, Bunsen burner flame, size of beaker, distance from Bunsen to gauze mat, thermometer used, position of Bunsen burner under beaker. **b** Height ball is dropped from, ruler used to measure bounce, surface ball is dropped on to, inflation of ball, type of ball, wear on ball **c** Length of string, mass of pendulum, angle of pull, stopwatch used to measure swing, type of string
- a** Independent: pulley system. Dependent: force. Controlled: spring balance, mass, cotton string, method of pulling the spring balance. **b** Independent: player. Dependent: time to complete a lap. Controlled: game console, game level, number of laps, time since eating. **c** Independent: different surfaces. Dependent: bounce height. Controlled: dropping the ball, superball, method of measuring bounce height.
- a** Independent: height. Dependent: time. Controlled: distance to run, time of day, warm up, shoes, clothing, type of ground
b Independent: string length. Dependent: amount of weight. Controlled: type of string, method of attaching weights, type of knot attaching string to support, amount mass increases by each time. **c** Independent: size of magnesium pieces. Dependent: time. Controlled: amount of acid, concentration of acid, size of test tube, cleanliness of magnesium, same stopwatch.

9 – Experimental controls

- [In order] carbon dioxide, electronic balance, bubbles, gas, 3.50 g, liquid, evaporation, water, 375 mL, 3.11 g, control, compare
- a** Seeds without light **b** Heart rate without exercise **c** Same food and exercise without diet bars **d** People with cancer not given the drug

10 – Accuracy and reliability

- a** inaccurate but reliable **b** accurate but unreliable **c** inaccurate and unreliable **d** accurate and reliable
- Examples include spring balance, weighing scales, thermometer, voltmeter
- a** yes **b** no
- a** 2.25 V, 2.0 V, 1.75 V **b** voltmeter 2

- 5 **a** yes **b** possibly
 6 **a** 50 m tape measure **b** bathroom scales **c** electronic balance, triple beam balance **d** stopwatch, electronic timer
 7 **a** yes **b** no
 8 **a** Z and P **b** Y – 1°C, Z – 0.1°C, P – accurate to the nearest 1 mm, Q – accurate to the nearest 1 cm
 9 **a** yes **b** yes
 10 Examples include: change in humidity, ruler slips while measuring, leaving a solution to boil for too long

11 – Validity

- 1 For example: people wouldn't be able to rely on scientist's results, drugs may hurt or kill people because the dosage was wrong, buildings and other structures might collapse and kill people because they weren't tested properly, people could get sick or die because food quality control was inadequate or wrong.
 2 Yes, because many people who like brussels sprouts also wear glasses and many people who don't like brussels sprouts don't wear glasses.
 3 See whether there is a correlation with the age of the people surveyed and wearing glasses or liking brussels sprouts. Chose a random sample of people who are not related to each other. Larger sample size.
 4 No, because there has not been any measurement of whether people are wearing contact lenses.
 5 **a** Independent: covered with glass/not covered with glass. Dependent: temperature **b** Size and thickness of shoebox, amount of sunlight/shading, time spent in the sun
 6 **a** Initial age of fish, size of tank, amount and type of food, position of tank, cleanliness of tank **b** Model of car, type of roads the cars travelled on, driver, weather, tyres
 7 No, because he doesn't know whether 34 tomatoes is what a plant would normally produce without fertiliser.
 8 Grow at least 20 tomato plants in exactly the same way but give only half of them fertiliser.
 9 Incorrect calibration, parallax error, incorrect use of measuring device, inappropriate measuring device, too large a limit of reading, uncontrollable error

12 – Quick quiz B

- 1 **a** Independent: temperature. Dependent: Vitamin C content. Controlled: juice from the same oranges, container size and material, amount of juice, amount of indicator, indicator **b** Indicator with water (no Vitamin C) is an experimental control to better judge colour change. **c** 0.1 mL **d** [in order] inaccurate, inaccurate, inaccurate, accurate **e** Use a dropper where the volume of each drop is known. Use a thermometer to measure the temperature of the juice in each location. Use a colour chart or a computerised device to determine the colour. **f** No, because most techniques were inaccurate. **g** Repeat the entire experiment at least ten times and see whether the results are similar. **h** She controlled all variables other than independent and dependent, such as the amount of orange juice, the amount of indicator, the type of container, the oranges. There is no reason to think she is not measuring Vitamin C content.
 2 1 Fill 20 identical vases with 250.0 mL of distilled water using a measuring cylinder. 2 Add 10.0 g of sugar to ten of the vases. Label these vases 'sugar'. 3 Stir each vase gently 15 times in a clockwise direction with a stirring rod. 4 Place all the vases in the same area of the same room. 5 Add one cut flower of the same species, development and time since being cut to each vase. 6 Each day, record the appearance of each flower.

13 – Describing measurements A

Quantity	Quantity's Symbol	Standard Unit	Standard Unit's Symbol
Length	l	metre	m
Mass	m	kilogram	kg
Time	t	second	s
Electric current	I	Amperes	A
Temperature	T	Kelvin	K
Luminous intensity	I _v	candela	cd
Amount of substance	n	mole	mol

- 2 **a** Mesopotamians, Egyptians, Indus Valley civilisations, Persians, French. **b** They were different from person to person and between different regions. **c** Easier to use in calculations. Can be reproduced consistently anywhere on Earth. **d** How far light travels in a vacuum in 1/299 792 458 of a second.

14 – Describing measurements B

- 1 Frequency, f , s⁻¹, Hertz, Hz. Energy, E , kgm²s⁻², Joule, J. Pressure, p , kgm⁻¹s⁻², pascal Pa
 2 **a** Acceleration, time, m/s over s = m/s × 1/s = m over s² = m/s² = ms⁻² **b** kg × ms⁻² = kgms⁻² **c** kgms⁻² × m = kgm²s⁻² **d** kg × (m/s)² = kgm²s⁻²
 3 m/s over m = m/s × 1/m = 1/s = s⁻¹ = units of frequency

15 – Scientific notation

- 1 **a** 5.4 × 10⁴ **b** 6.7 × 10¹⁰ **c** 7.800 × 10³ **d** 2.356 41 × 10⁸
 2 **a** 4.5 × 10⁻⁴ **b** 9.0 × 10⁻² **c** 3 × 10⁻¹¹ **d** 3.687 × 10⁻³
 3 **a** 34 000 **b** 500 **c** 54 900 **d** 5 500 000
 4 **a** 0.000 34 **b** 0.05 **c** 0.054 9 **d** 0.000 005 50

16 – Significant figures

- 1 **a** 3 **b** 1 **c** 3 **d** 2 **e** 5 **f** 5 **g** 3 **h** 5
 2 **a** 0.000 011 5 or 1.15 × 10⁻⁵ **b** 6.32 **c** 0.02 or 2 × 10⁻² **d** 54.5 × 10⁻⁵ or 5.45 × 10⁻⁶ (better answer) **e** 5.6 × 10²
 3 **a** 5.0 × 10³ **b** 27 **c** 1.5 × 10² **d** 0.006 9 or 6.9 × 10⁻³ **e** 4.37
 4 **a** 0.009 **b** 88.81 **c** 2 372 **d** 23 **e** 0.8 m **f** 23 **g** 3.3 × 10² **h** 4.47 × 10⁻¹¹
 5 **a** 0.004 0 **b** 6 × 10¹ **c** 1.0 × 10¹² **d** 1.67 × 10⁻²⁴ **e** 0.32
 6 **a** 23 cm **g** 0.13 s

17 – Orders of magnitude

Prefix	Symbol	Multiplies by	Order of magnitude	Meaning
tera	T	× 1 000 000 000 000	10 ¹²	one trillion
giga	G	× 1 000 000 000	10 ⁹	one billion
mega	M	× 1 000 000	10 ⁶	one million
kilo	k	× 1 000	10 ³	one thousand
deci	d	× 0.1	10 ⁻¹	one tenth
centi	c	× 0.01	10 ⁻²	one hundredth
milli	m	× 0.001	10 ⁻³	one thousandth
micro	μ	× 0.000 001	10 ⁻⁶	one millionth
nano	n	× 0.000 000 001	10 ⁻⁹	one billionth
pico	p	× 0.000 000 000 001	10 ⁻¹²	one trillionth

kilo	No prefix	deci	milli
54 km	= 54 000 m	= 540 000 dm	= 54 000 000 mm
0.000 453 9 kL	= 0.453 9 L	= 4.539 dL	= 453.9 mL
0.000 16 kV	= 0.16 V	= 1.6 dV	= 160 mV
0.000 2 kA	= 0.2 A	= 2 dA	= 200 mA
0.001 kg	= 1 g	= 10 dg	= 1 000 mg

- 3 4 – 0.002 kilograms. 1 – 20 000 grams. 5 – 20 milligrams.
 3 – 2 000 centigrams. 2 – 0.002 megagrams

- 4 **d** 10^1 **e** 10^5 **f** 10^{-11} **g** 10^{-3} **h** 10^7 **i** 10^{-5} **j** 10^{-5}
 5 10^9 The Sun's diameter – 1.4 Gm, 10^7 Earth's diameter – 12.8 Mm, 10^3 Mt Everest – 8 848 m, 10^2 The Eiffel Tower – 321 m, 10^0 A human – 1.7 m, 10^{-2} A queen honeybee – 1.5 cm, 10^{-3} A raindrop – 1.5 mm, 10^{-6} A bacterium – 2 μ m, 10^{-7} A virus – 100 nm, 10^{-10} A hydrogen atom – 100 pm

19 – Quick quiz

- 1 **a** 6.544×10^4 **b** 3.20×10^{-5} **c** 4.50×10^{-7}
 2 **a** 2 **b** 2 **c** 3 **d** 3
 3 **a** 5.46×10^6 (better answer) or 54.6×10^5 **b** 9.9×10^{-7} **c** 3.5×10^3
 4 **a** 74 **b** 1.49×10^{25} **c** 5.9×10^5 **d** 2.0×10^2 m
 5 kilo – 10^3 – k, mega – 10^6 – M, micro – 10^{-6} – μ , milli – 10^{-3} – m
 6 **a** 10^8 **b** 10^{-8} **c** 10^{-3} **d** 10^7 **e** 10^5 **f** 10^{12}
 7 **a** 10^{-1} to 10^{-2} **b** 10^{-3} to 10^{-4} **c** 10^1 **d** 10^6

Number without prefixes	Number with prefix	Scientific notation	No. of significant figures
9 007 m	9.007 km	9.007×10^3 m	4
0.581 m	58.1 cm	5.81×10^{-1} m	3
0.000 012 g	12 μ g or 0.012 mg	1.2×10^{-5} g	2

20 – Drawing tables

- 1 Rule 1 lines, Rule 2 mistakes, Rule 3 column, Rule 4 data, Rule 5 heading, Rule 6 same, Rule 7 variable, Rule 8 not

2

Flavour	Frequency / Number of people
Cola	8
Lemonade	4
Raspberry	10
Lime	3
Orange	5

3

Element	Electrical conductivity	Colour	Malleability
copper	conducts	orange	malleable
silicon	conducts	grey	brittle
sulfur	doesn't conduct	yellow	brittle
carbon	conducts	black	brittle
iron	conducts	silver	malleable
aluminium	conducts	silver	malleable

4

Measuring cylinder	Beaker
Used to measure liquids to the nearest millilitre	Used to measure approximate amounts of liquids
Found in science laboratories	Found in science laboratories
Tall and thin	Short and wide
Any similarity - Made of glass, have volume scale, have pouring lip	Made of glass, have volume scale, have pouring lip

5 **a**

	Time (s)			
Liquid	Trial 1	Trial 2	Trial 3	Average
Olive oil	2.36	2.25	2.40	2.34
Water	0.45	0.40	0.50	0.45
Honey	20.09	11.47	13.33	14.96
Ethanol	0.12	0.33	0.27	0.24

b Honey, because it took the longest to slide down the ramp.

21 – Drawing timelines

- 1 Rule 1 equally, same. Rule 2 most. Rule 3 pencil. Rule 4 names, brief, diagrams. Rule 5 line. Rule 6 large, once

23 – Scientific models

- 1 **Mathematical** - d, **Computer simulation** - c, **Diagrammatic** - f, **Structural or scale models** - a, **Thought experiments** - b, **Descriptive** - g, **Analogies** - e
 3 Examples include: prevented from driving a car, cause injuries, afraid to go to public places or swimming.
 4 Examples include: poorer grades at school, harder to concentrate at work so can't earn as much money, difficulty reacting in time when crossing roads or driving, forgetting important things or not being bothered to do them.
 5 1 Microscope images of nerve cells, 2 Diagram of how nerve cells recycle materials, 3 Identification of important protein, 4 3D computer simulation of protein shape, 5 'Fitting' potential blocking drugs into the protein's active binding site, 6 Using rats to test potential drugs for toxicity and effectiveness, 7 Testing potential drugs for epilepsy in humans
 6 Diagrammatic, computer simulation, analogy (animal model is like humans) (A second analogy is used in the passage, but not by the researchers)
 7 Helping people live better lives without the effects of a disease. Better understanding of nerve cell processes which could lead to other discoveries.
 8 Potential drugs may all be toxic and they have to start again, drugs that are toxic to rats may be fine for humans, microscope images may not show enough detail for a good diagrammatic model, computer simulation may be using incorrect assumptions.
 9 Benefits: relative shape and size of parts can be seen
 Limitations: not actual size, doesn't show how the cell works, not 3D like actual cell, connections with other cells not shown

24 – Using a database – designing

- 1 For example: with each person concentrating on their job, it can be done faster and more work can be completed OR with each person concentrating on their job, it can be done with fewer mistakes and the work is done better.
 2 Examples include: Doctors' surgery, local council, hospital, library, school
 3 Examples include: species name, habitat, food sources, number of individuals, predators, colour, type of organism
 4 Examples include: isotopes, electrical conductivity, atomic radius, colour, hardness, electron configuration

25 – Using a database – entering data

- 1
- | Temperature ($^{\circ}$ C) | Temperature (K) |
|-----------------------------|-----------------|
| 25 | 298 |
| -90 | 183 |
| 1 067 | 1 340 |
| -186 | 87 |
- 2 Absolute zero is the lowest possible temperature. It is 0K or -273.15 $^{\circ}$ C. At absolute zero an object has no heat energy and all atomic motion ceases.

26 – Using a database – retrieving data

- 1 C
 2 A – Excel, B – Word or Publisher, D – Powerpoint

27 – Quick quiz

- 1 **a** Form **b** Design view **c** Record **d** Wizard **e** Objects **f** Query **g** Field **h** Primary key **i** Report **j** Data **k** Database **l** Table
 2 **a** true **b** false **c** true **d** false
 3 **a** If the power goes out, changes are already saved. Don't have to keep pressing 'save' for every record.
b If a mistake is made, any previous overwritten data is lost.
 4 Examples include: name, street address, suburb, home phone number, mobile phone number, email address

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108 Main Rd
Clayton South
VIC 3169
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